

Integrated Circuit ECE326

Lec. 14

Layout

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Outlines



- ☐ Stick diagram
- ☐ Exercice_01
- ☐ Exercice_02
- ☐ Exercice_03

Outlines



- ☐ **Stick diagram**
- ☐ **Exersice_01**
- ☐ **Exersice_02**
- ☐ **Exersice_03**

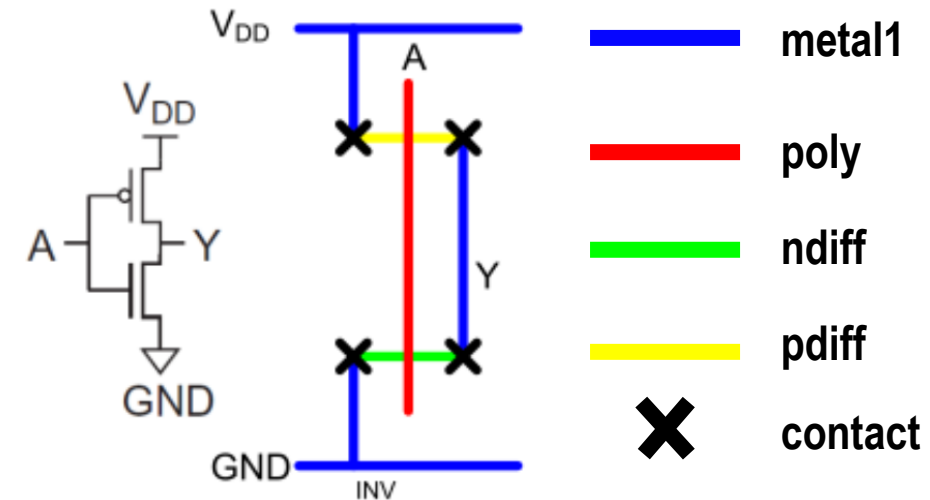
Stick diagram (1/2)



- ❑ A stick diagram is a symbolic layout to plan layout **quickly**.
- ❑ Wires are drawn as stick figures with **no width, so no need to scale**
- ❑ It is always much **faster** to design layout on paper using stick diagram **first** before using the layout CAD tool

❑ Important Tips (Do)

- ❑ Metal1 **GND** rail at **bottom**
- ❑ Metal1 **V_{DD}** rail at **top**
- ❑ Metal1 makes **connections**
- ❑ **Horizontal** N-diffusion and p-diffusion strips
- ❑ **Vertical** polysilicon gates
- ❑ Transistor = intersection of poly and diffusion
- ❑ nMOS usually are placed near **GND**, and pMOS near **V_{DD}**
- ❑ **Must** connect ndiff to metal and then metal to pdiff (No direct connection)



Does not show more information about masks of the layout

Stick diagram (2/2)



- ❑ Step1:- Construct the logic diagram of the given function (pull-up and pull-down networks)
- ❑ Step2:- Identify each **Node** between transistors and give them a unique name (Main Nodes: V_{DD} , O/P, GND and Sub-Nodes: 1,2,3...)
- ❑ Step3:-Draw the **Euler diagram**.
- ❑ Step4:-Identify the **Euler path** for pull-up and pull-down networks
 - ❑ **Nodes can be repeated** but **Paths cannot be repeated**.
 - ❑ The path should be **valid** in both Pull-up and pull-down networks.
- ❑ Step5:-Draw the stick diagram
 - ❑ Align V_{DD} , O/P, GND (**Horizontal lines**)
 - ❑ Align pdiff and ndiff (**Horizontal lines**)
 - ❑ Align Euler path (**Vertical lines**)
 - ❑ Identify the main nodes and sub-nodes (X)
 - ❑ **Connect** the main Nodes and then Sub nodes.

Outlines

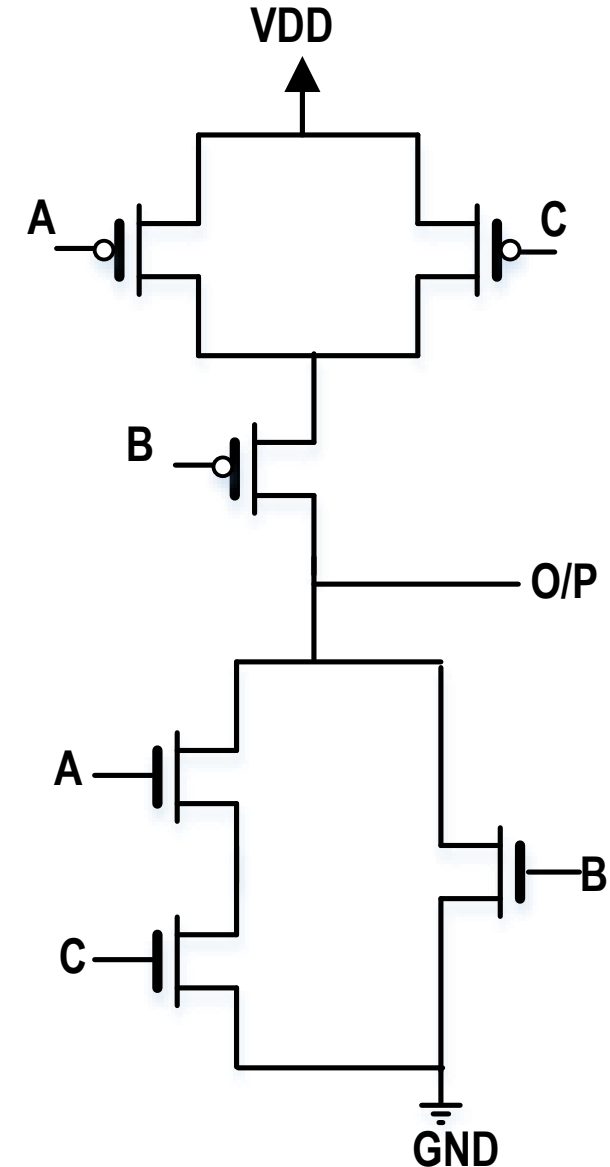


- ☐ Stick diagram
- ☒ Exercise_01
- ☐ Exercise_02
- ☐ Exercise_03

Exersice_01



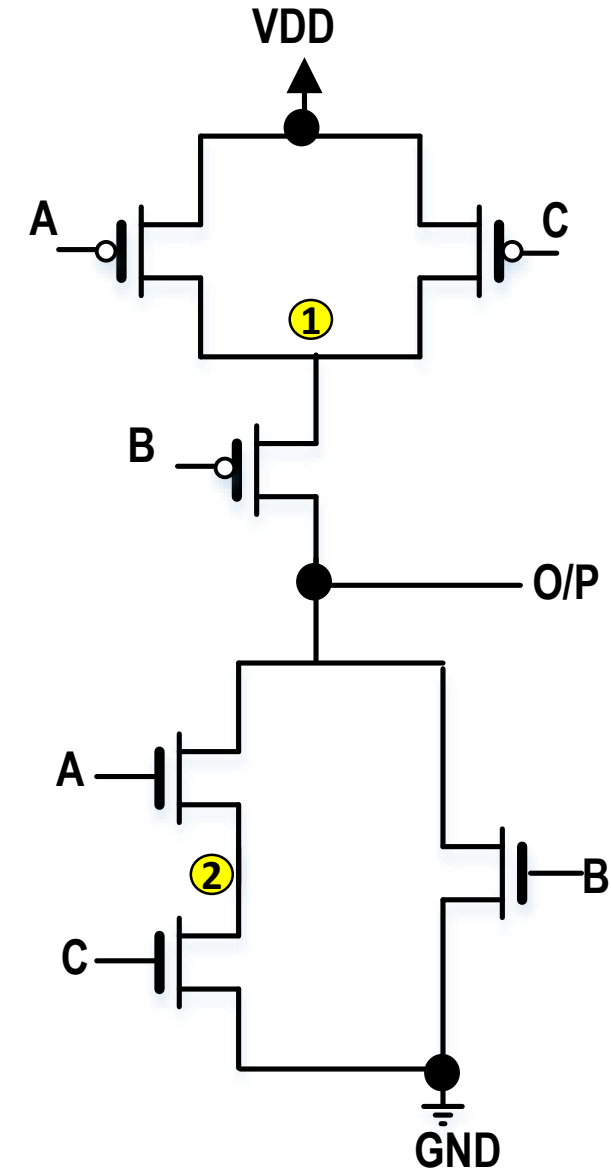
- Draw the stick diagram of the following circuit
- Steps1:- construct the logic diagram of the given function (pull-up and pull-down networks) → **given**



Exercise_01



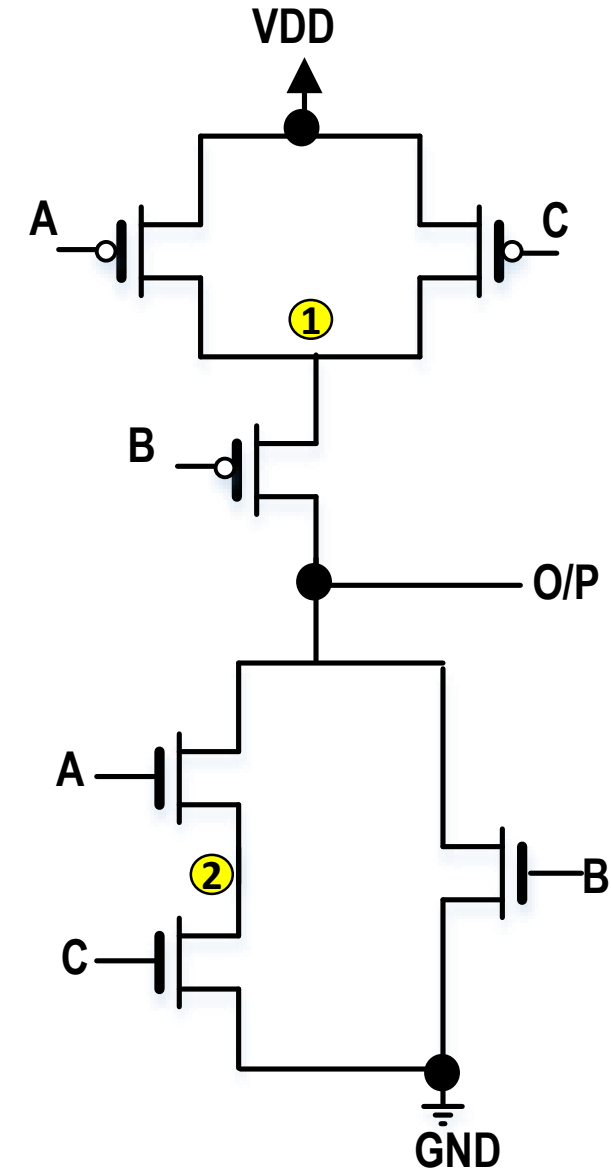
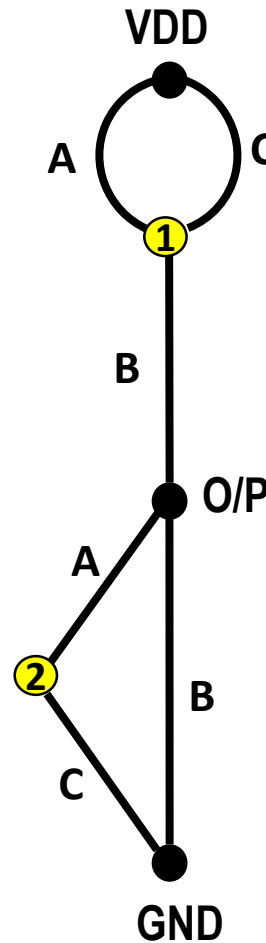
- ☐ Draw the stick diagram of the following circuit
- ☐ Step2:- Identify each connection between transistors only and give them a unique name.
 - ☐ Main Nodes: VDD, O/P, GND
 - ☐ Sub-Nodes: 1,2



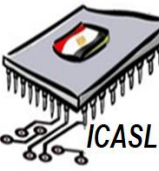
Exercise_01



- Draw the stick diagram of the following circuit
- Step3:- Draw the Euler diagram



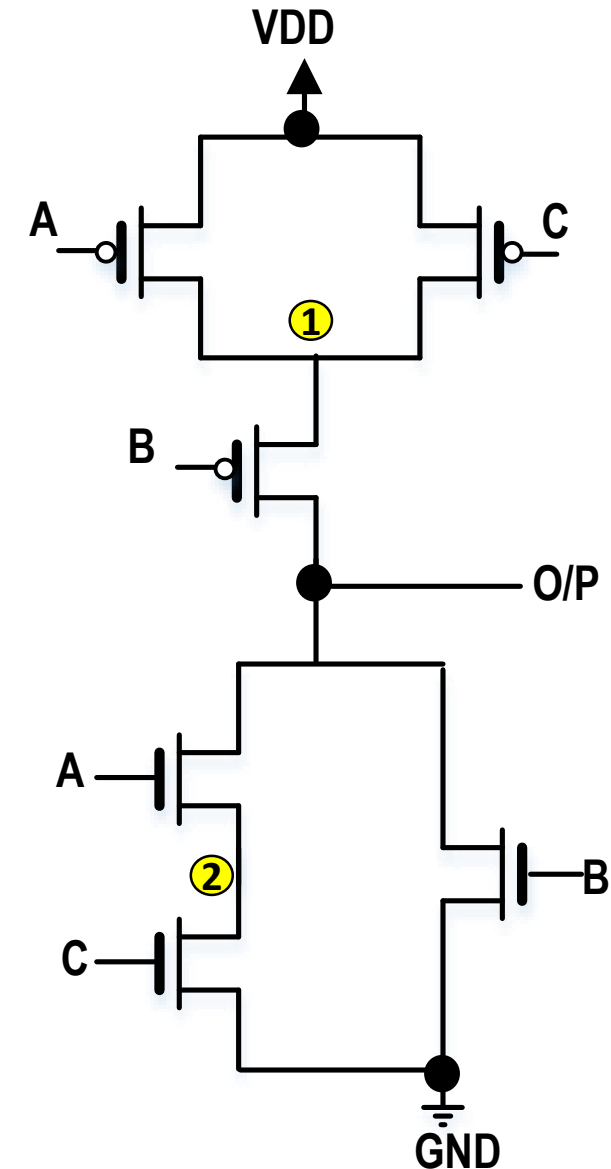
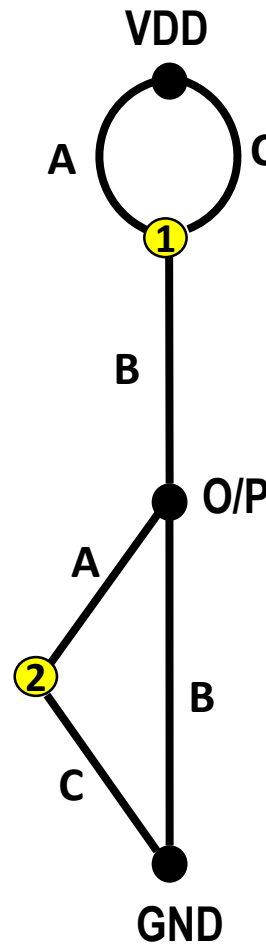
Exercise_01



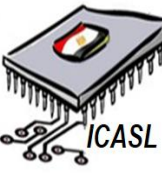
- Draw the stick diagram of the following circuit
- Step4:- Identify the Euler path

Condition of Euler Path

1. Nodes can be repeated but Paths cannot be repeated.
2. The path should be valid in both Pull-up and pull-down networks.



Exersice_01

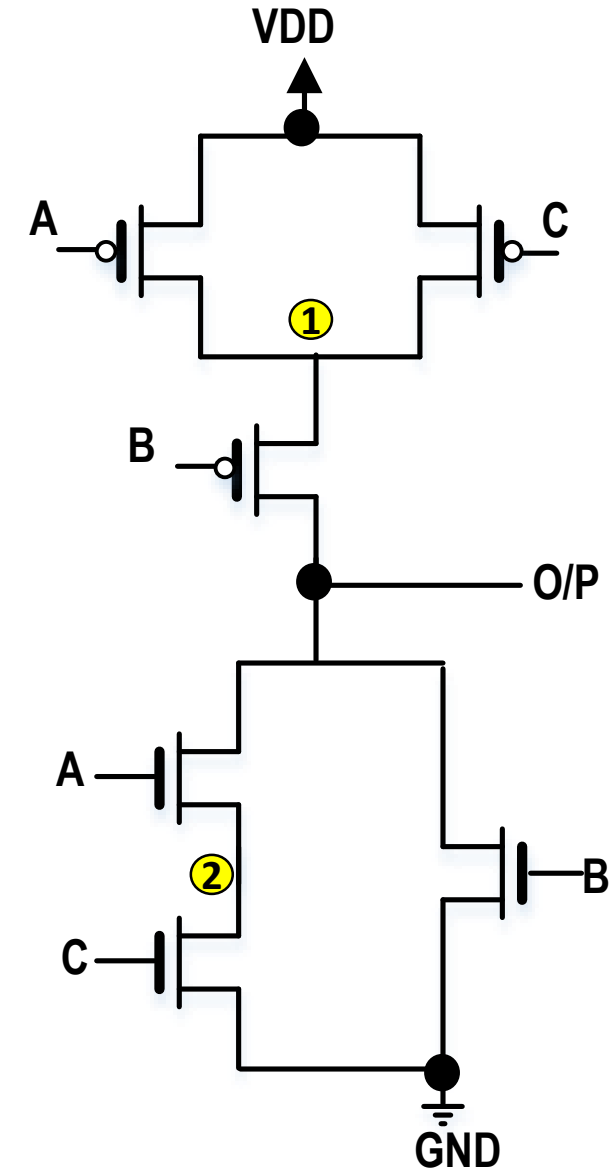
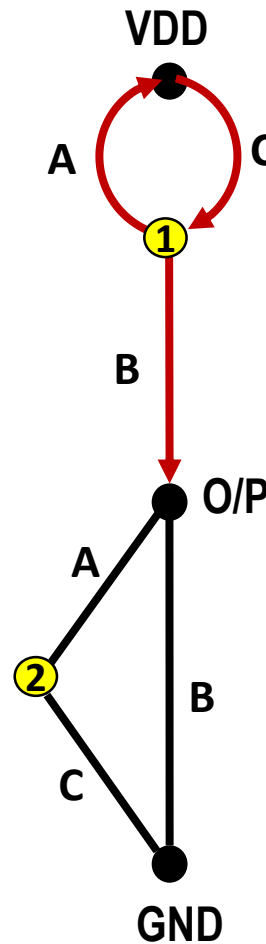


□ Draw the stick diagram of the following circuit

□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→C→B



Exercise_01



□ Draw the stick diagram of the following circuit

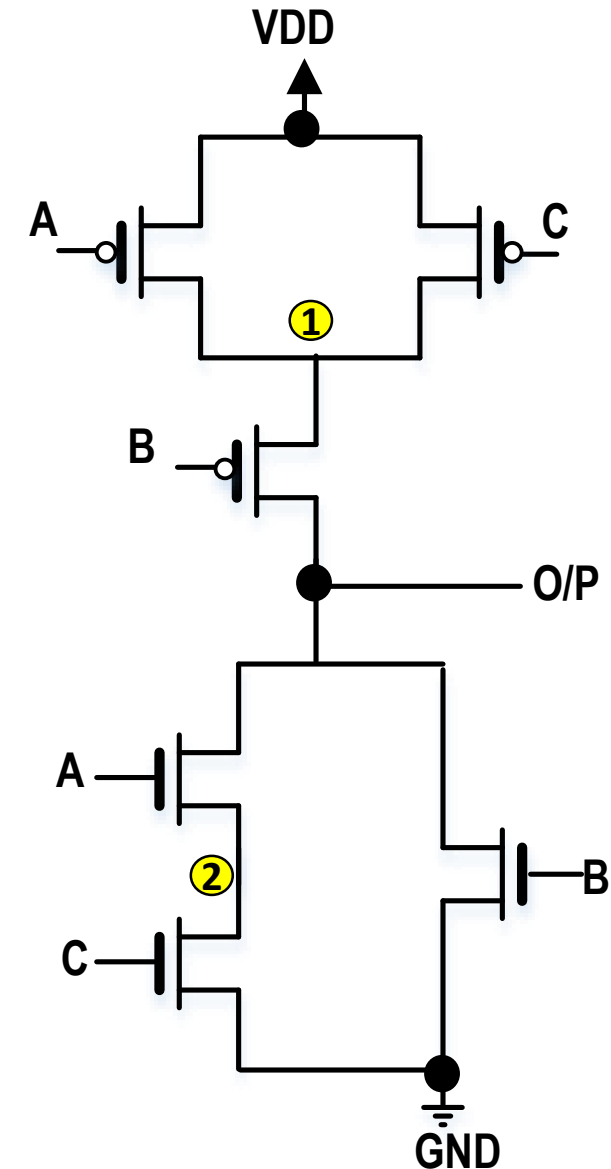
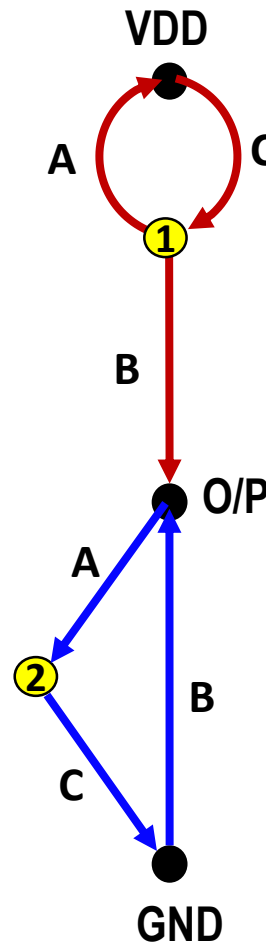
□ Step4:- Identify the Euler path

Euler path for Pull-up network is

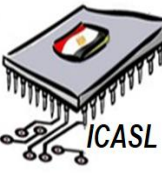
A→C→B

Euler path for Pull-down network is

A→C→B



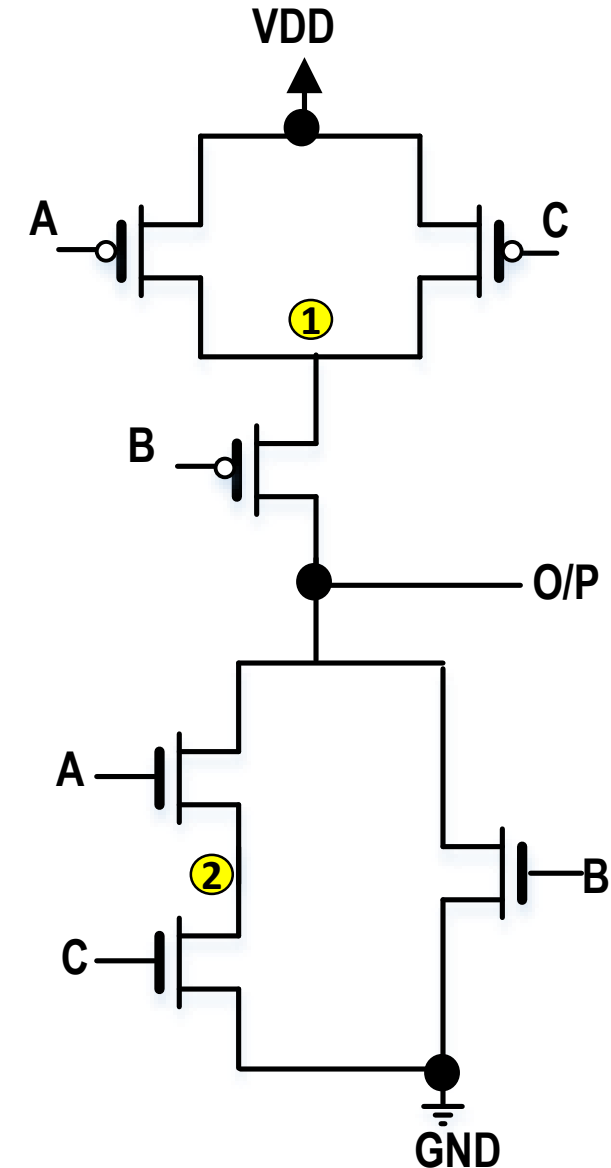
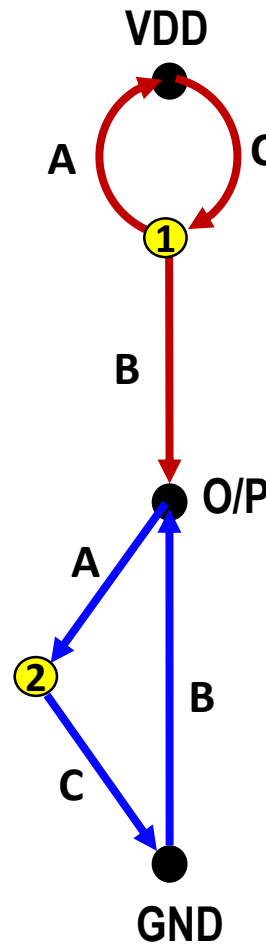
Exersice_01



□ Draw the stick diagram of the following circuit

□ Step5:- Draw the stick diagram

$A \rightarrow C \rightarrow B$ (in order)



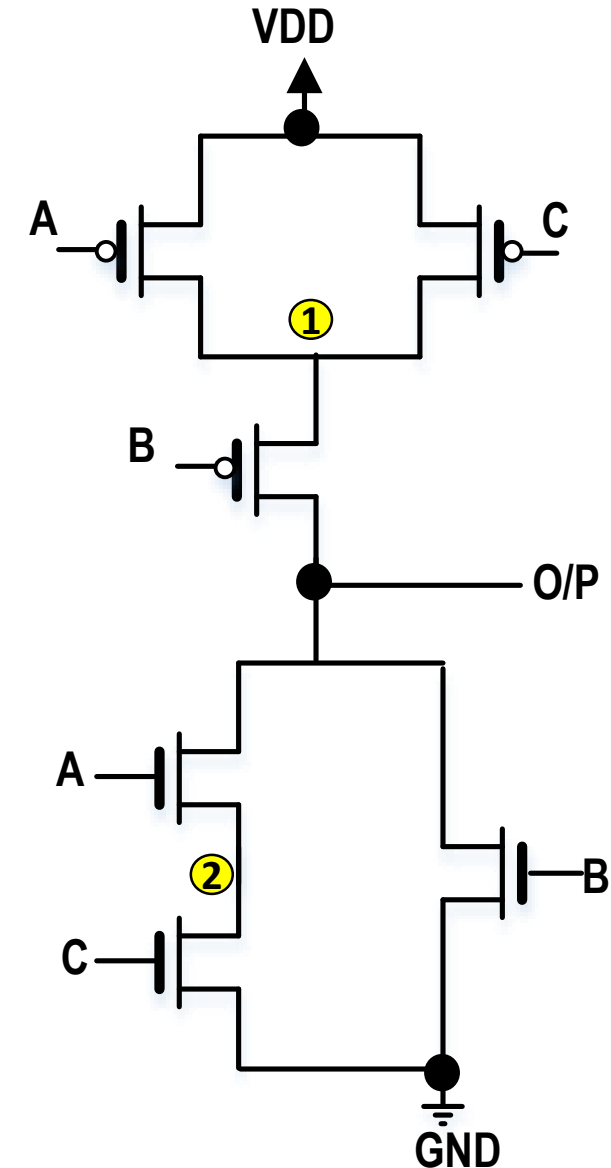
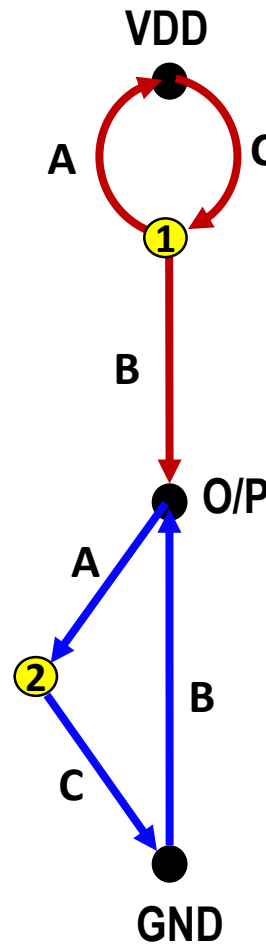
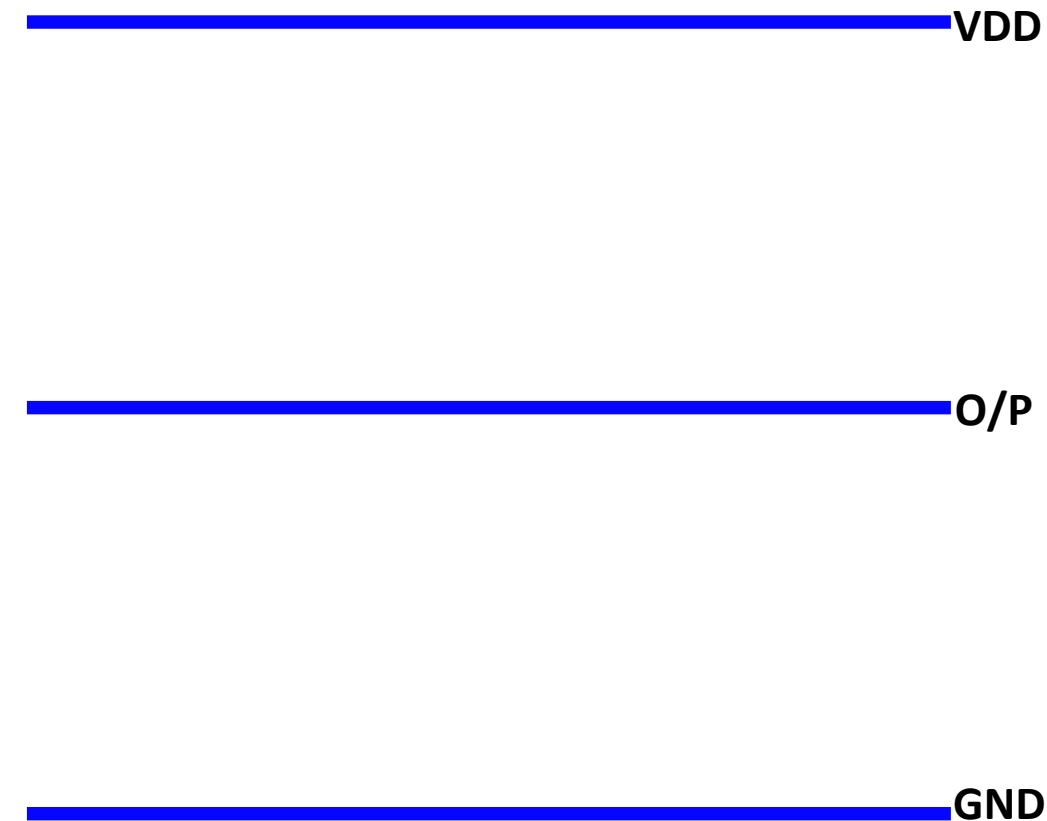
Exercise_01



☐ Draw the stick diagram of the following circuit

☐ Step5:- Draw the stick diagram

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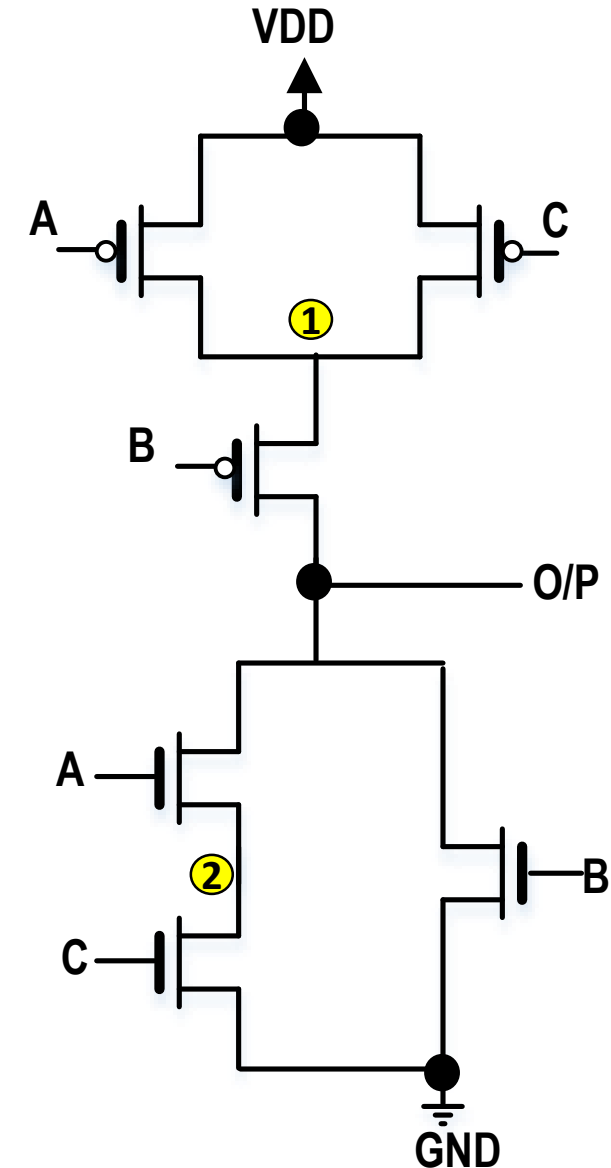
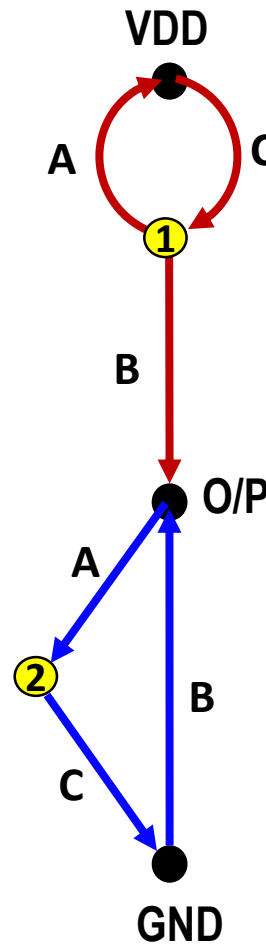
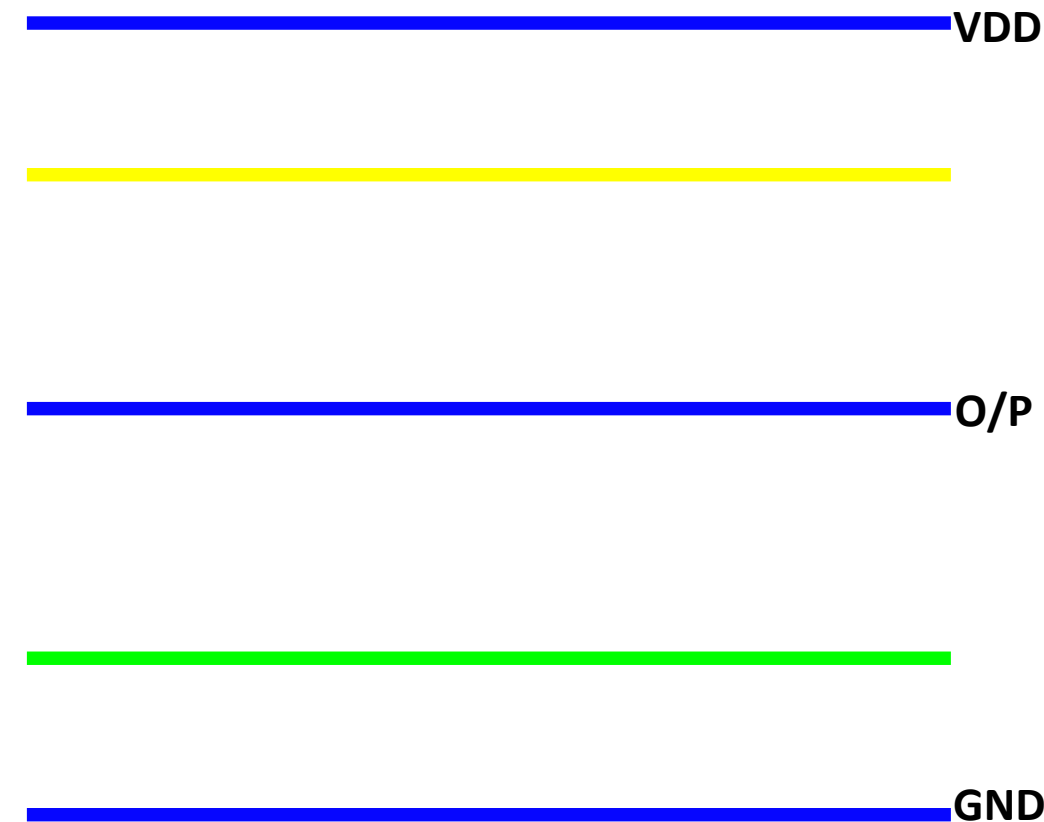
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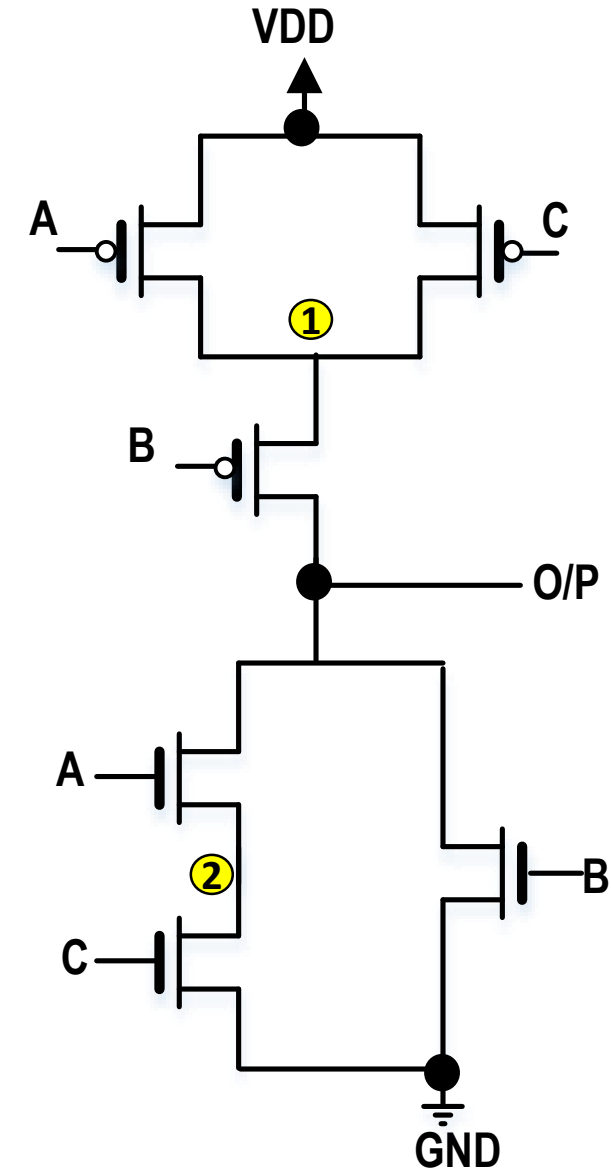
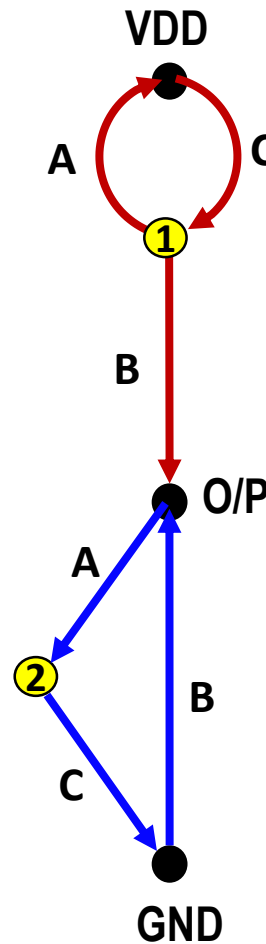
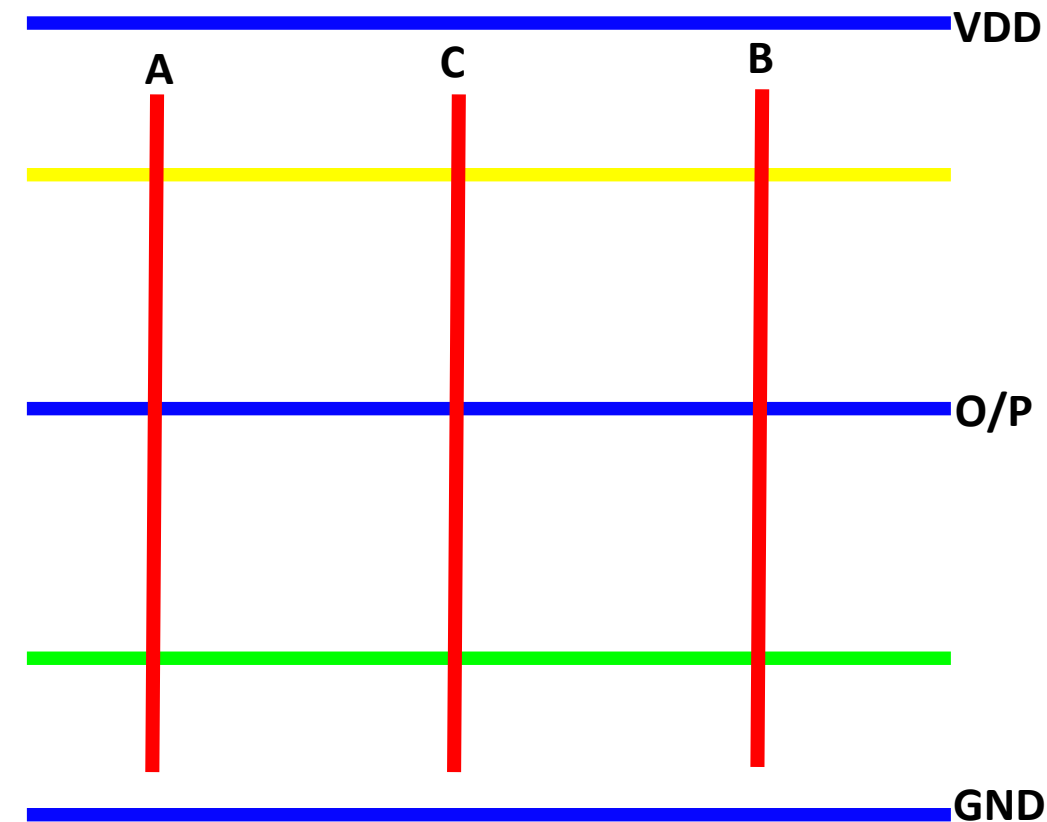
Exercise_01



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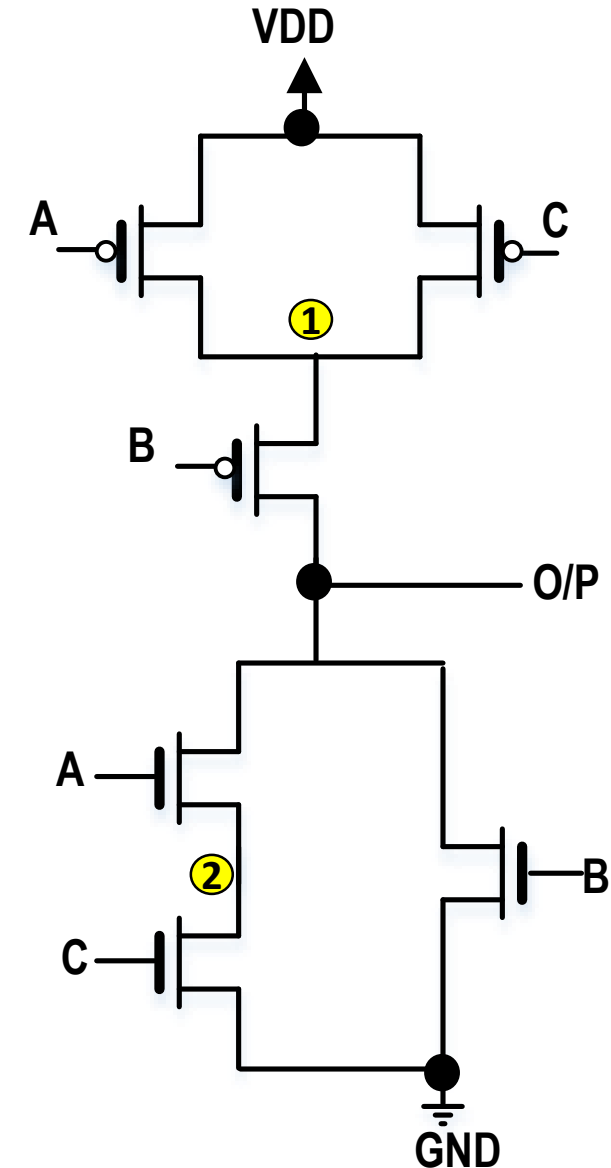
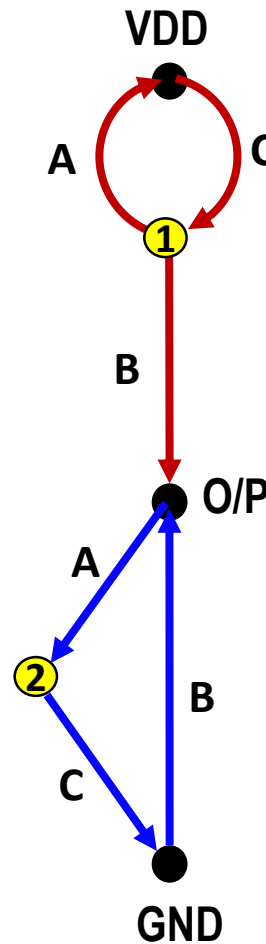
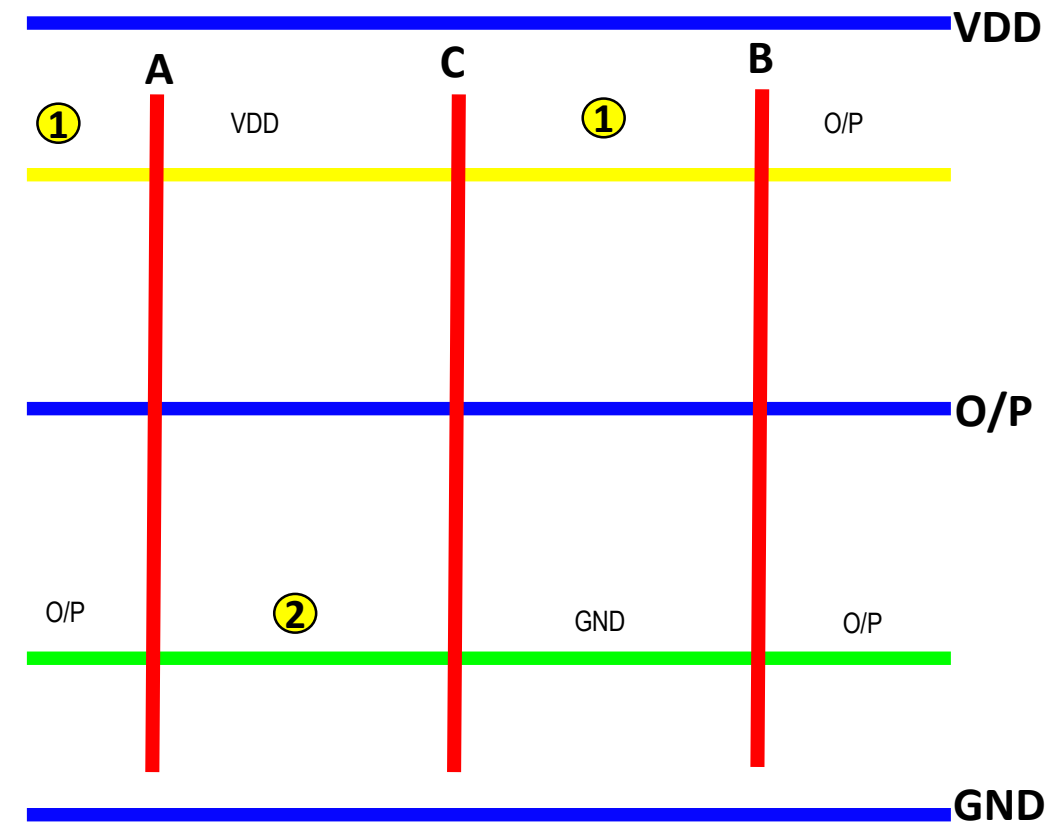
Exersice_01



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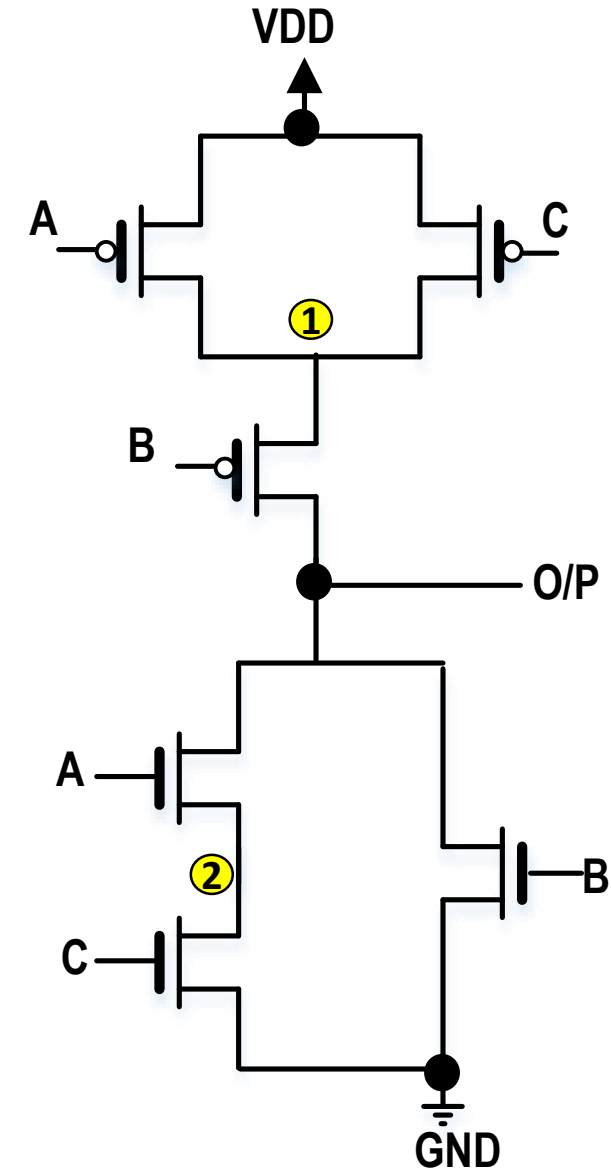
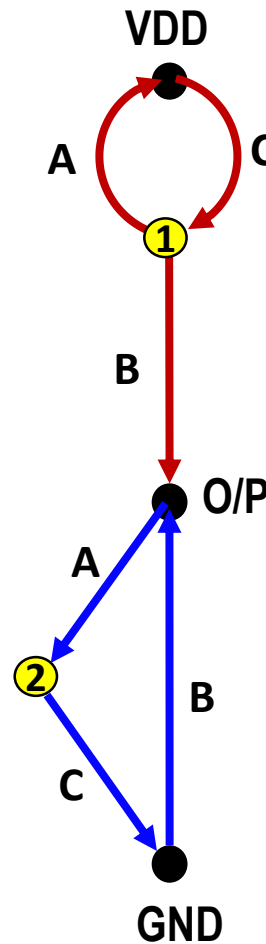
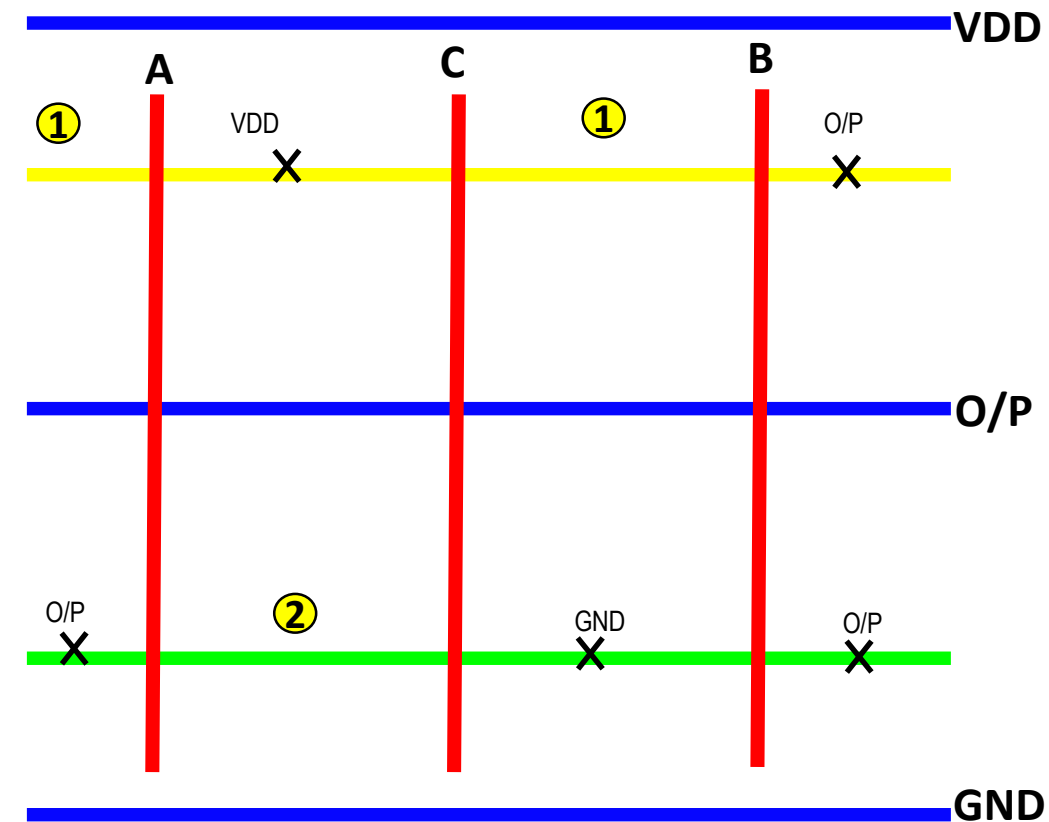
Exersice_01



□ Draw the stick diagram of the following circuit

□ Step5:- Draw the stick diagram

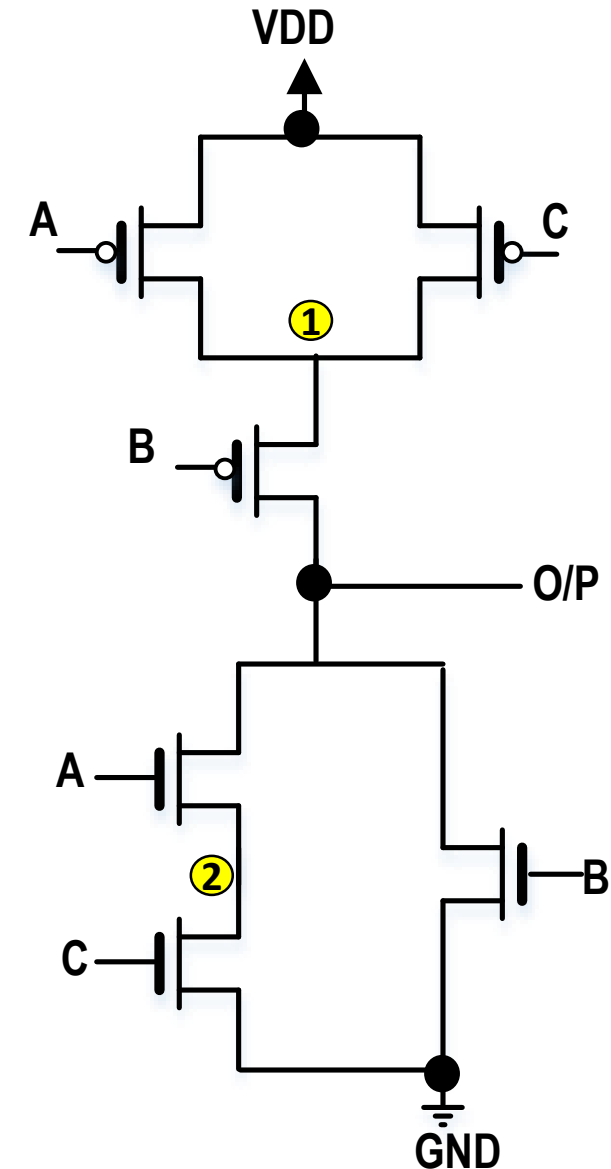
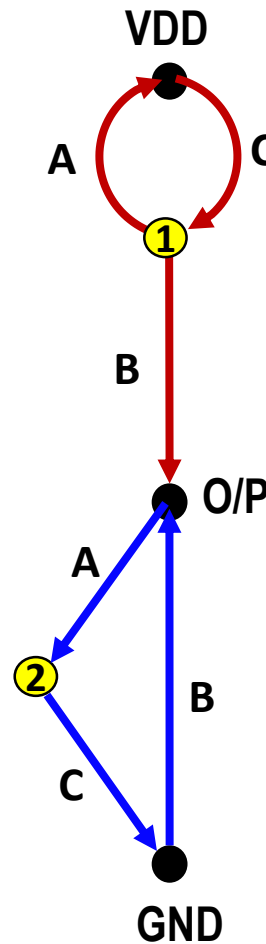
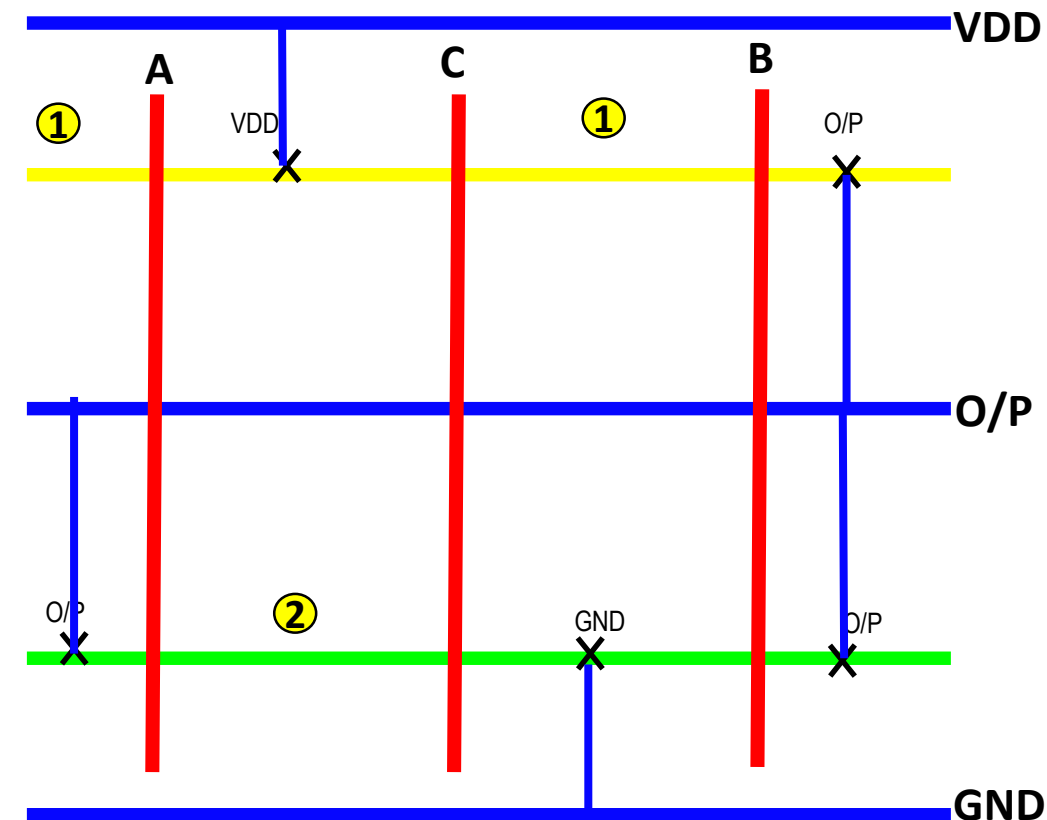
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Exercise_01



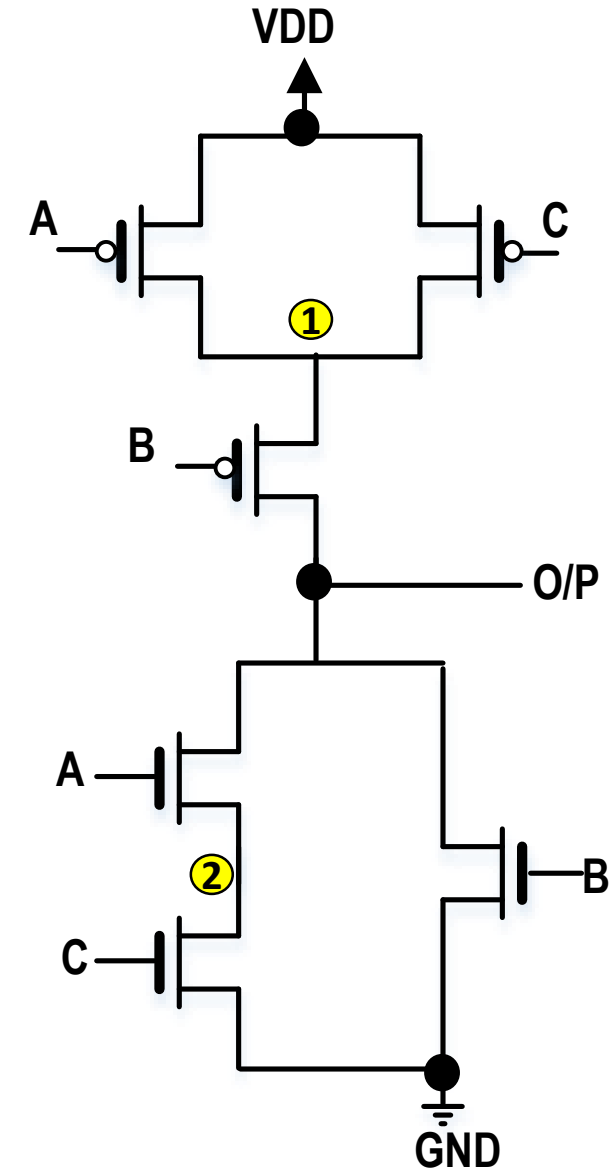
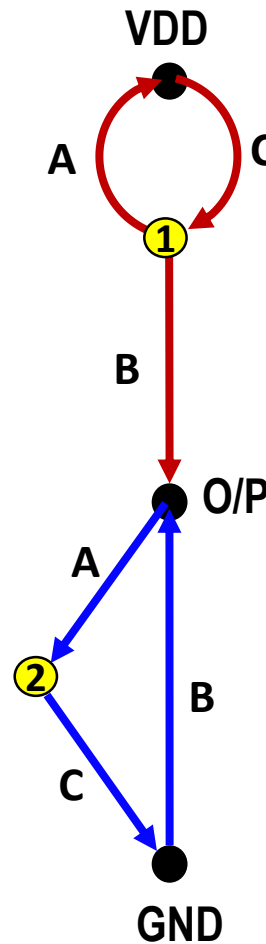
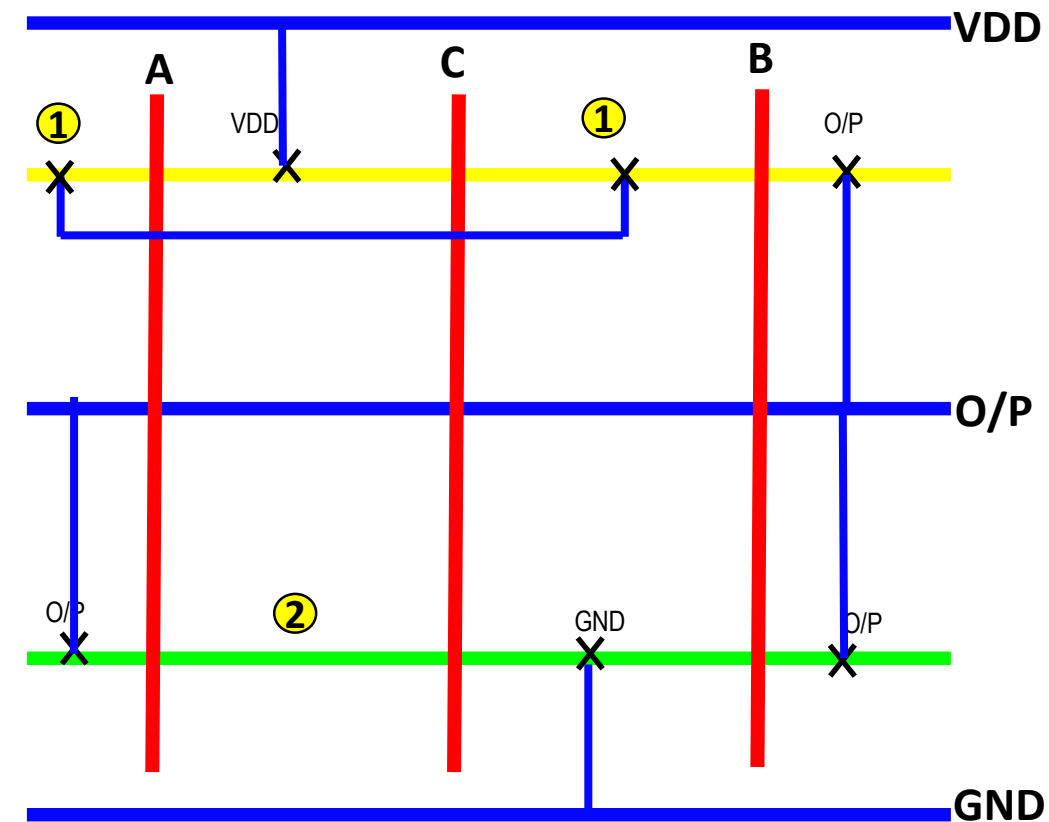
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 $A \rightarrow C \rightarrow B$ (in order)



Exersice_01



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 $A \rightarrow C \rightarrow B$ (in order)

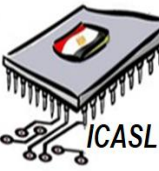


Outlines

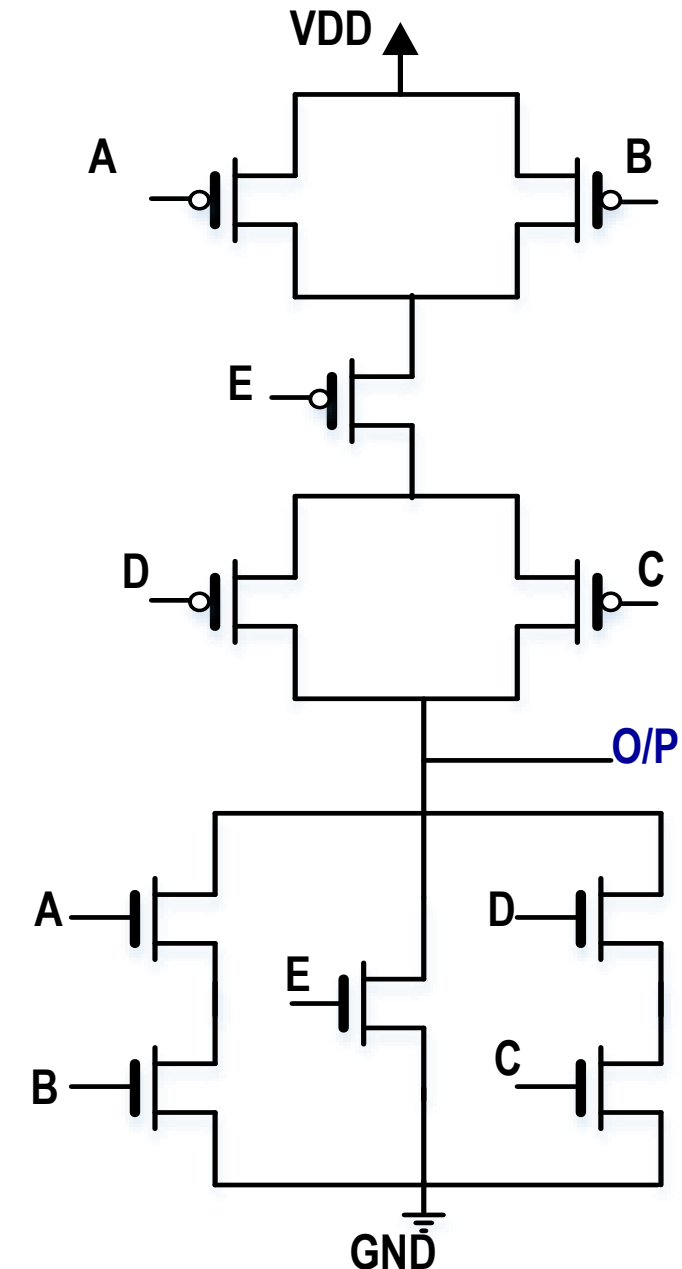


- ☐ Stick diagram
- ☐ Exercise_01
- ☒ Exercise_02
- ☐ Exercise_03

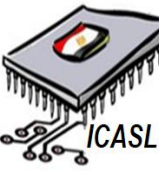
Exercise_02



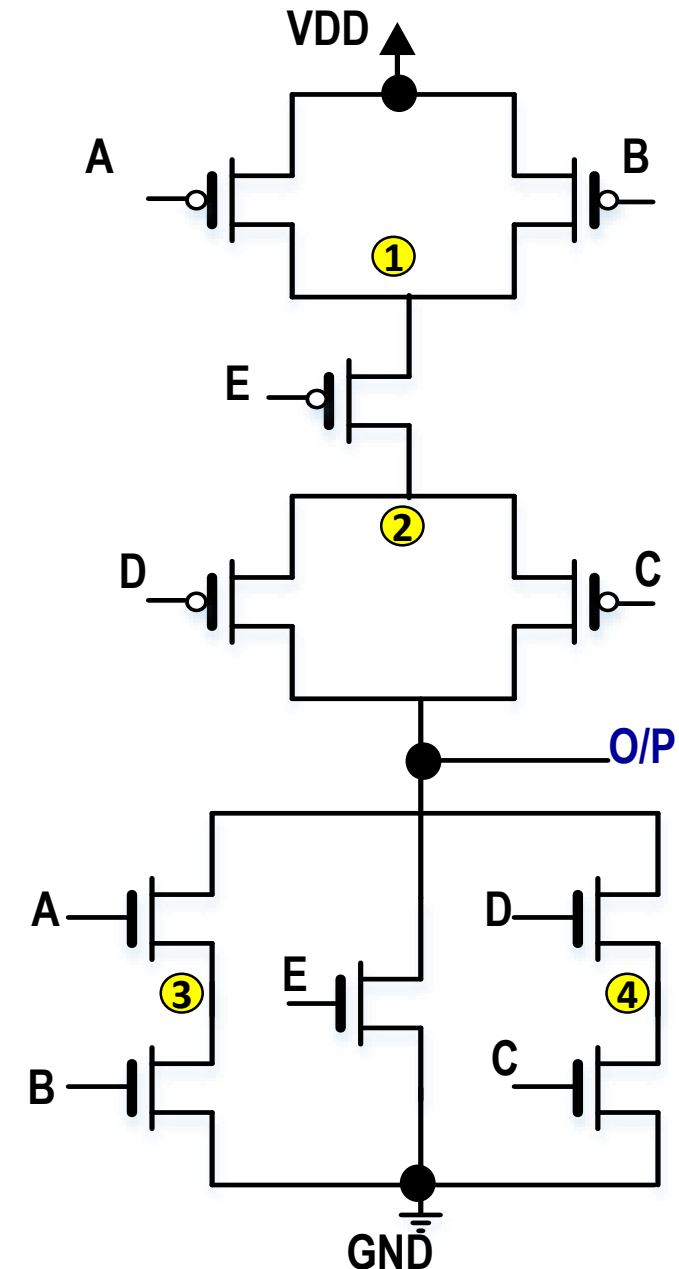
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- Steps1:- construct the logic diagram of the given function (pull-up and pull-down networks)→
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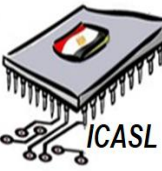
Exercise_02



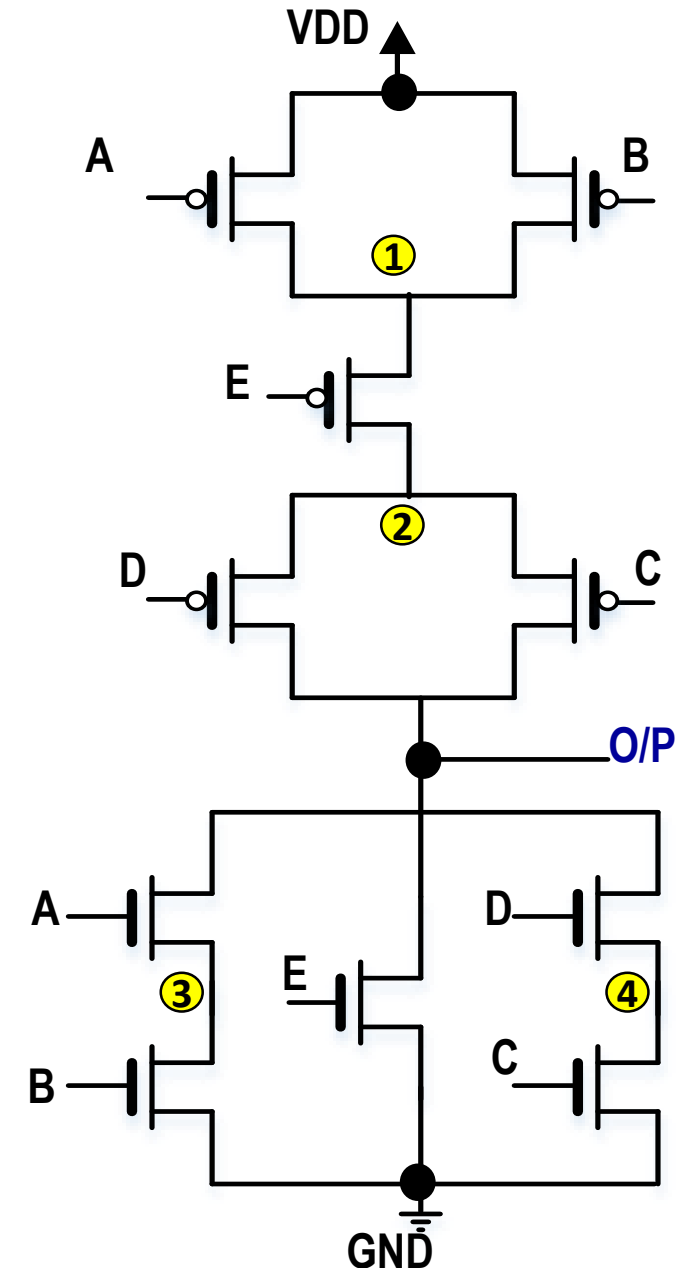
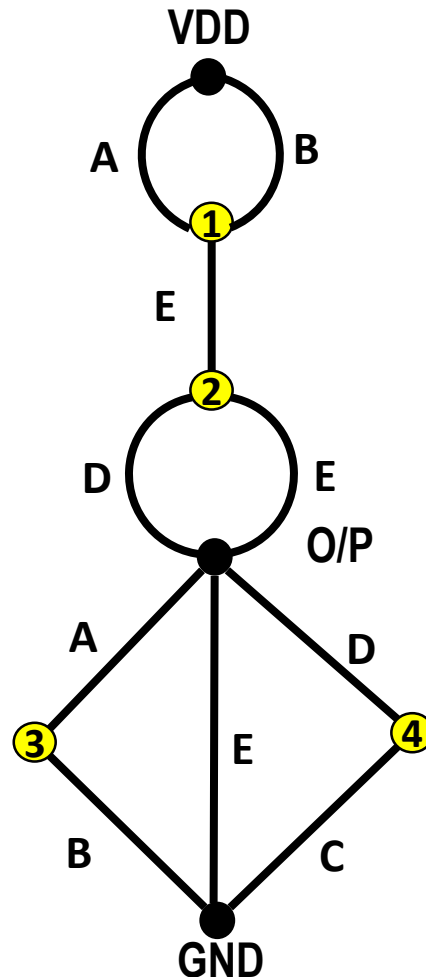
- ☐ Draw the stick diagram of the following circuit
- ☐ Step2:- Identify each connection between transistors only and give them a unique name.
 - ☐ Main Nodes: VDD, O/P, GND
 - ☐ Sub-Nodes: 1,2,3,4



Exersice_02

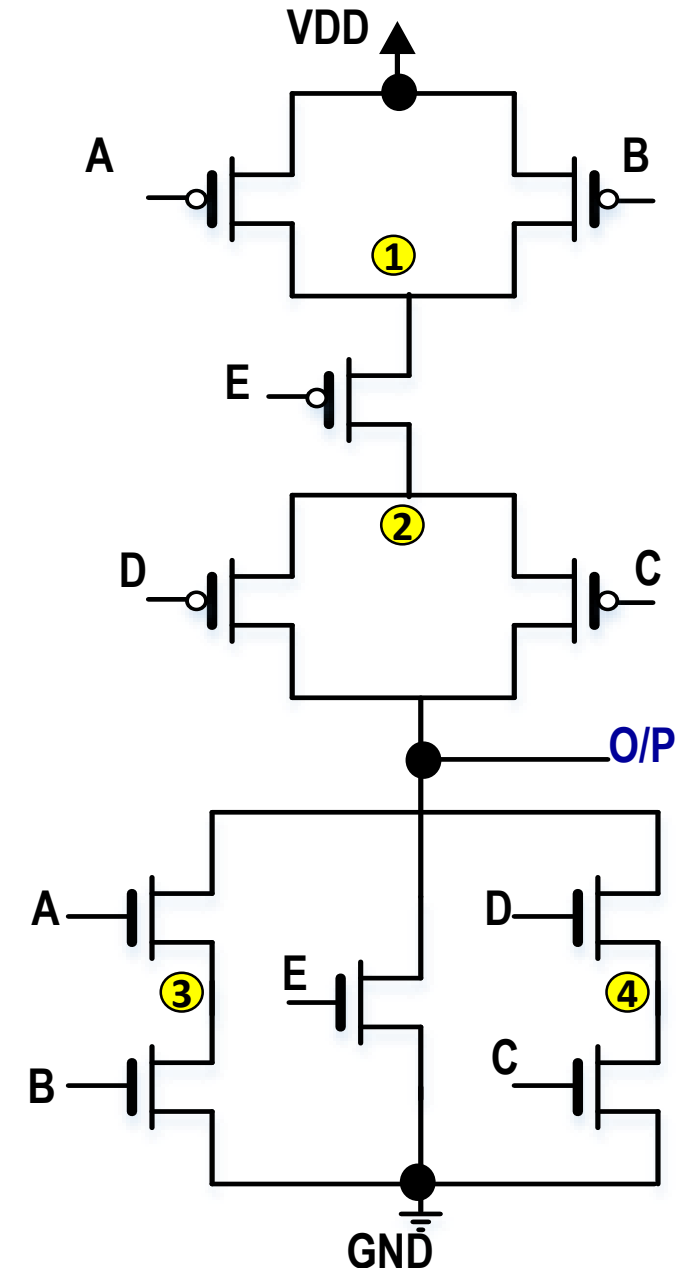


- Draw the stick diagram of the following circuit
- Step3:- Draw the Euler diagram.

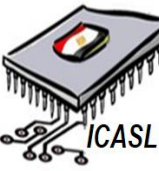


- ## Condition of Euler Path

Paths cannot be repeated



Exercise_02

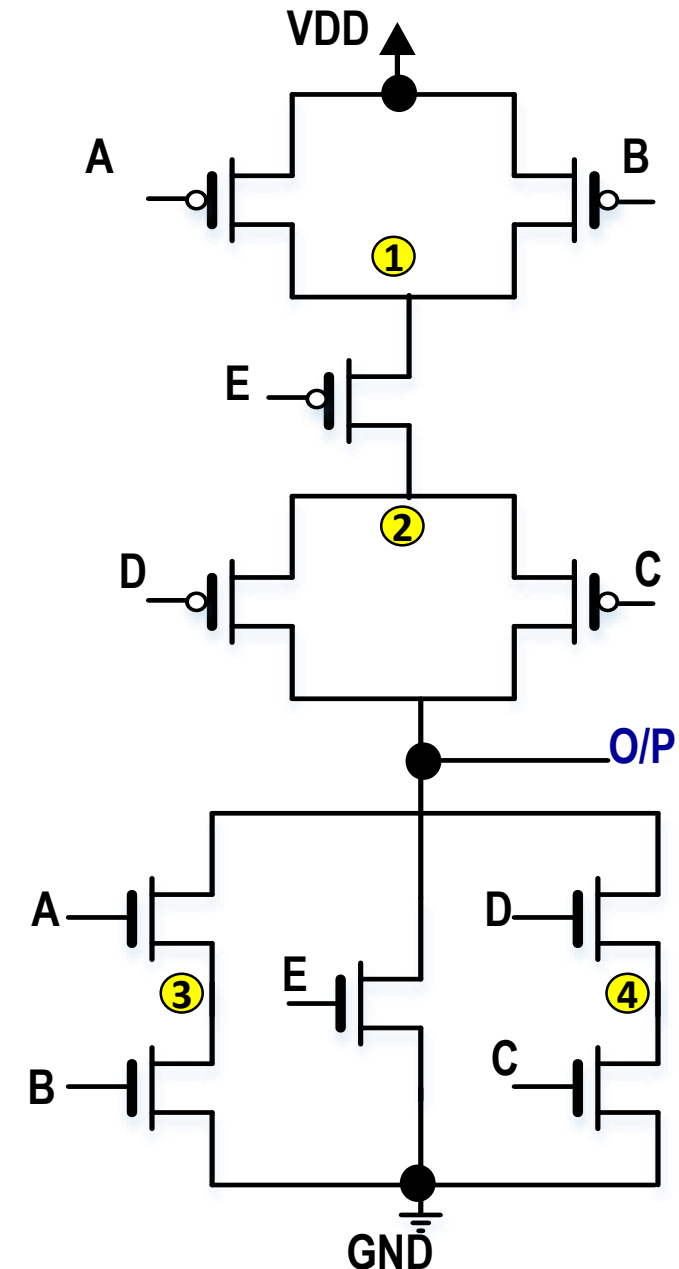
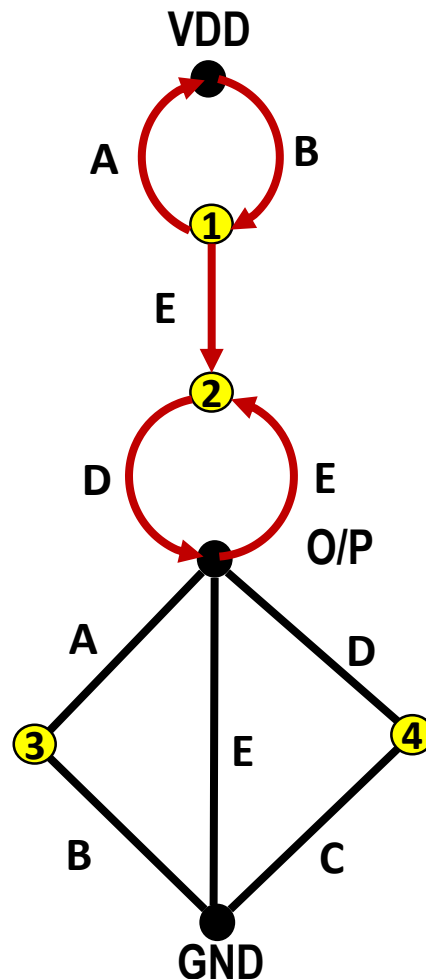


□ Draw the stick diagram of the following circuit

□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→B→E→D→C



Exersice_02



□ Draw the stick diagram of the following circuit

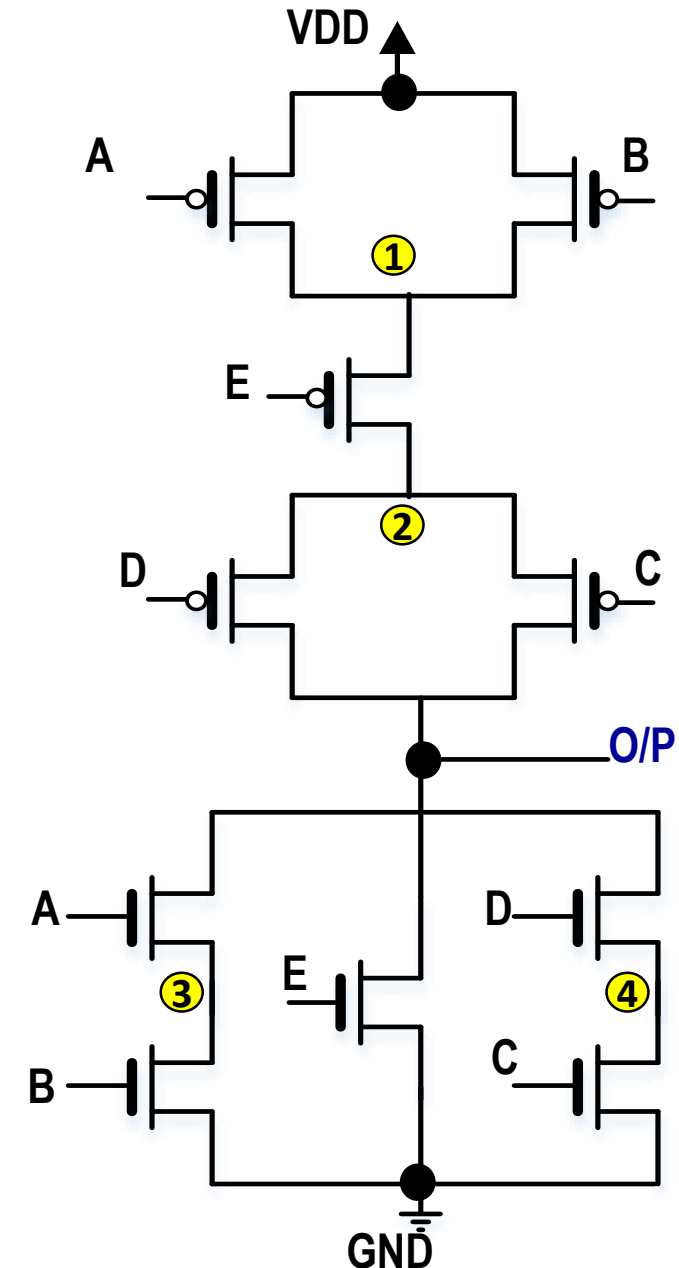
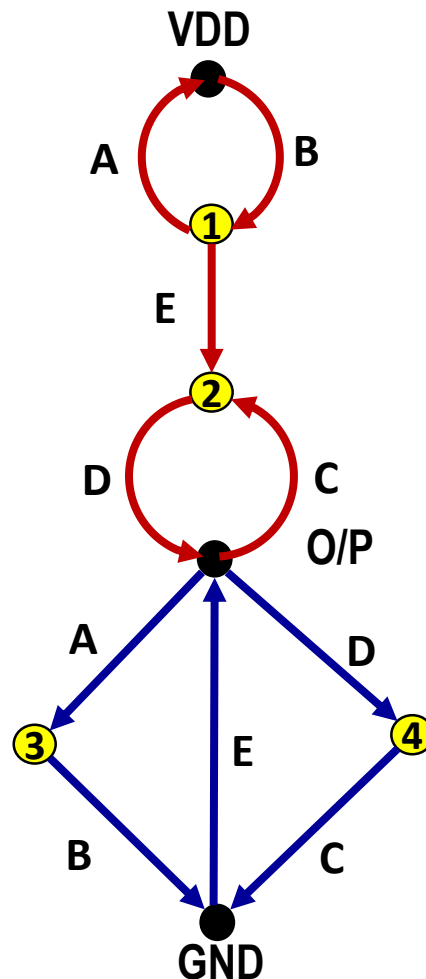
□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→B→E→D→C

Euler path for Pull-down network is

A→B→E→D→C



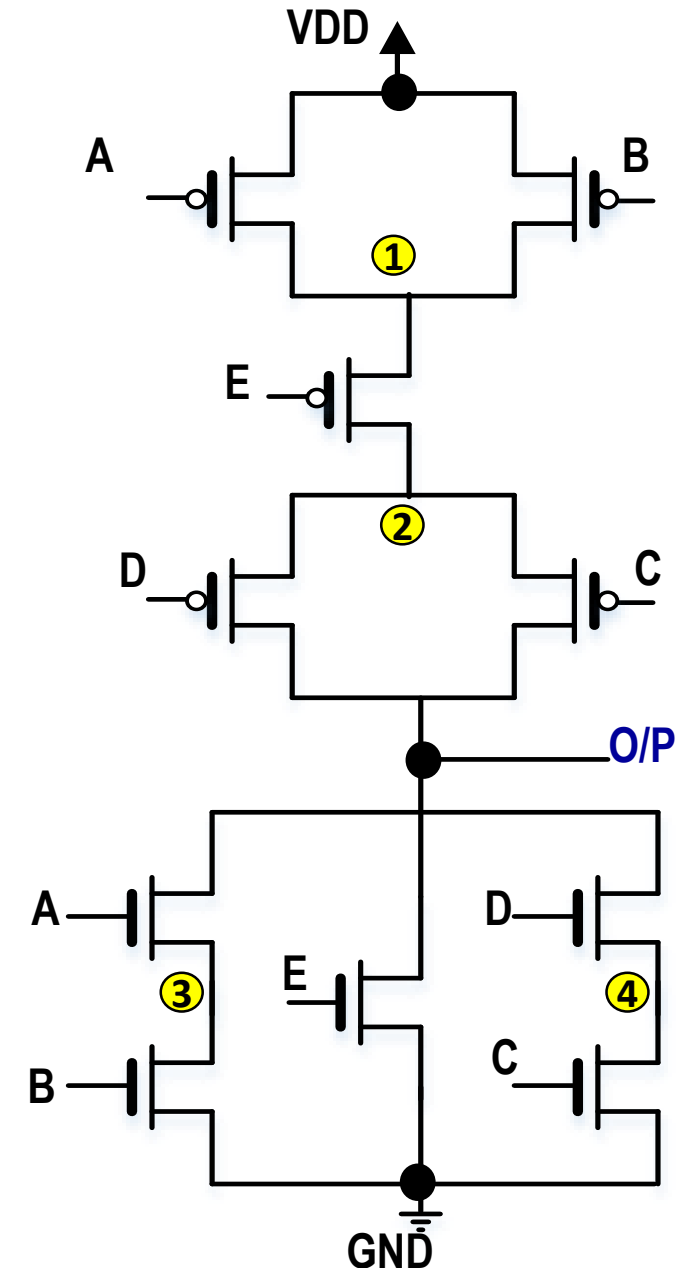
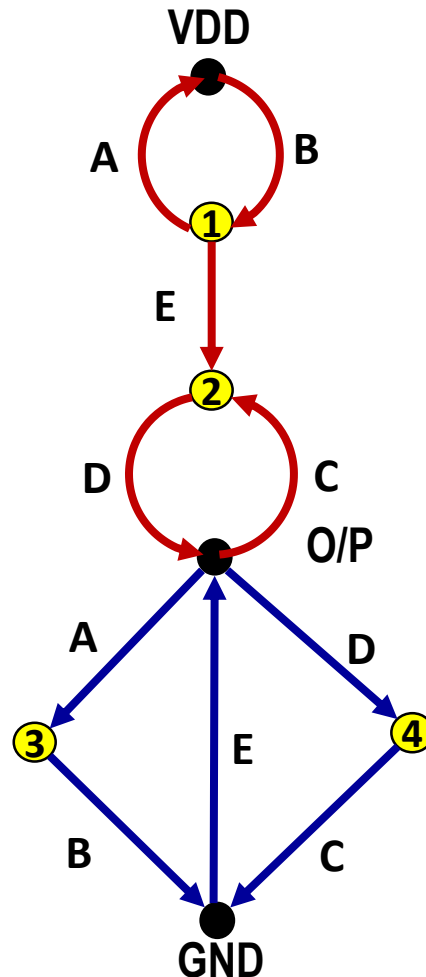
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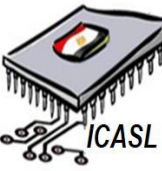
□ Draw the stick diagram of the following circuit

□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow E \rightarrow D \rightarrow C$ (in order)



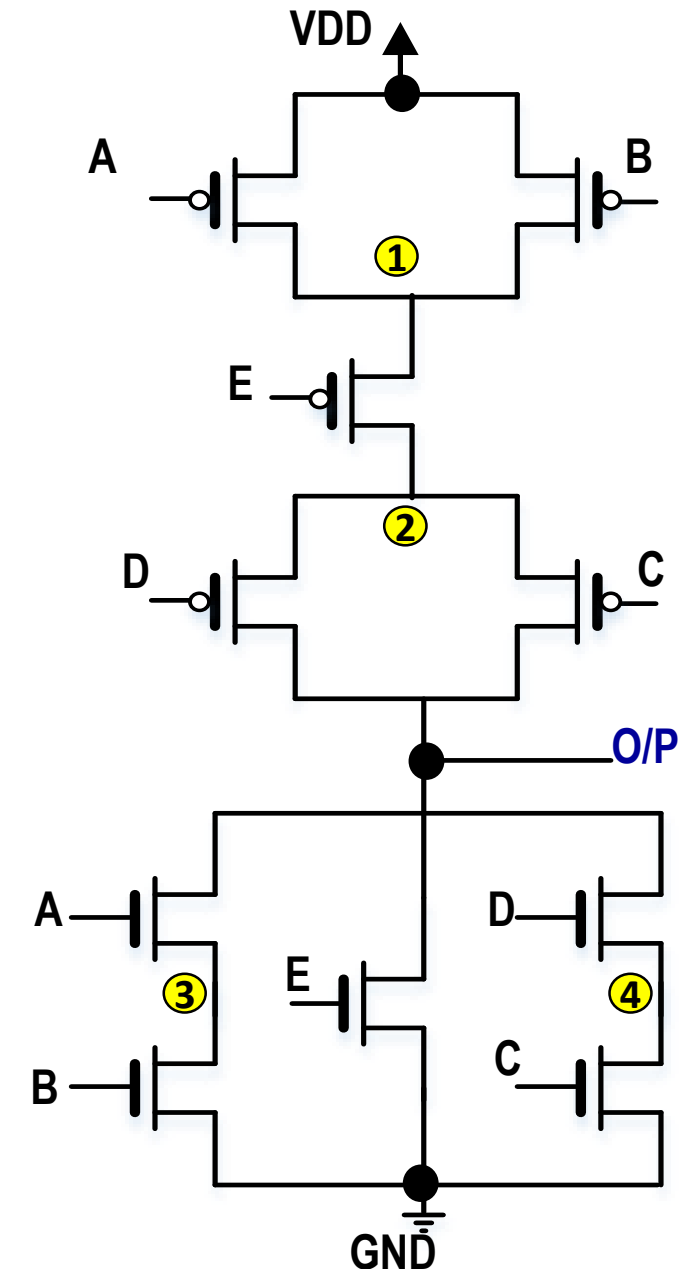
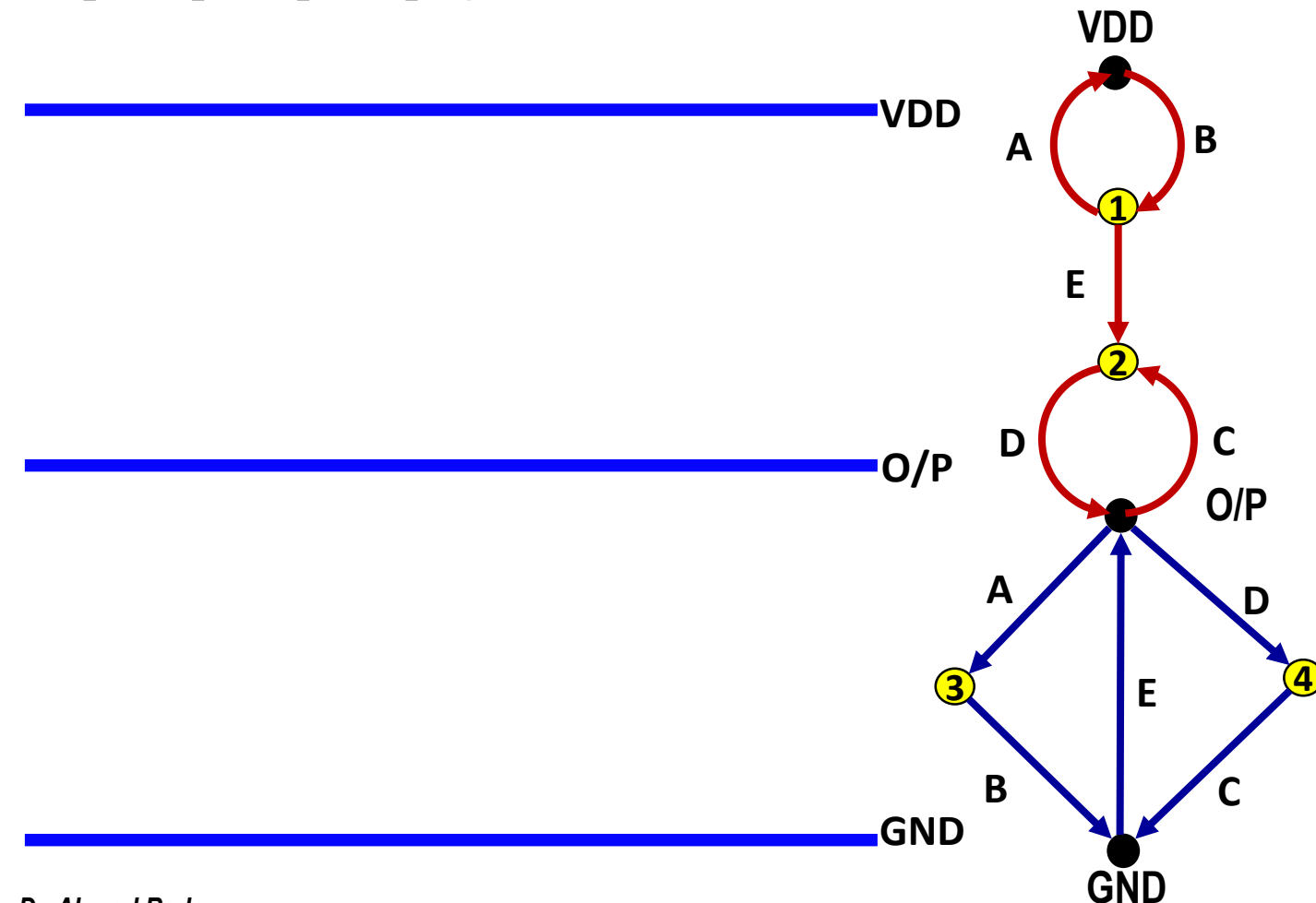
Exersice_02



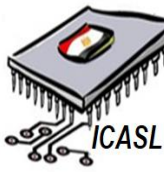
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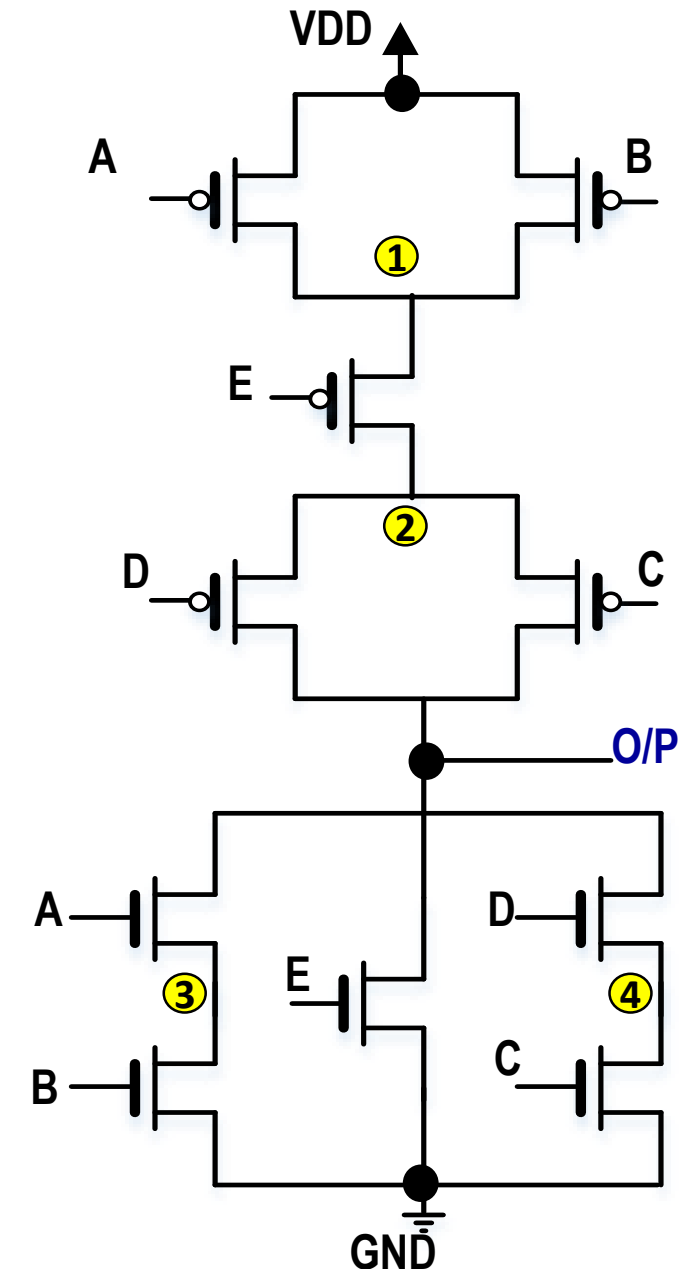
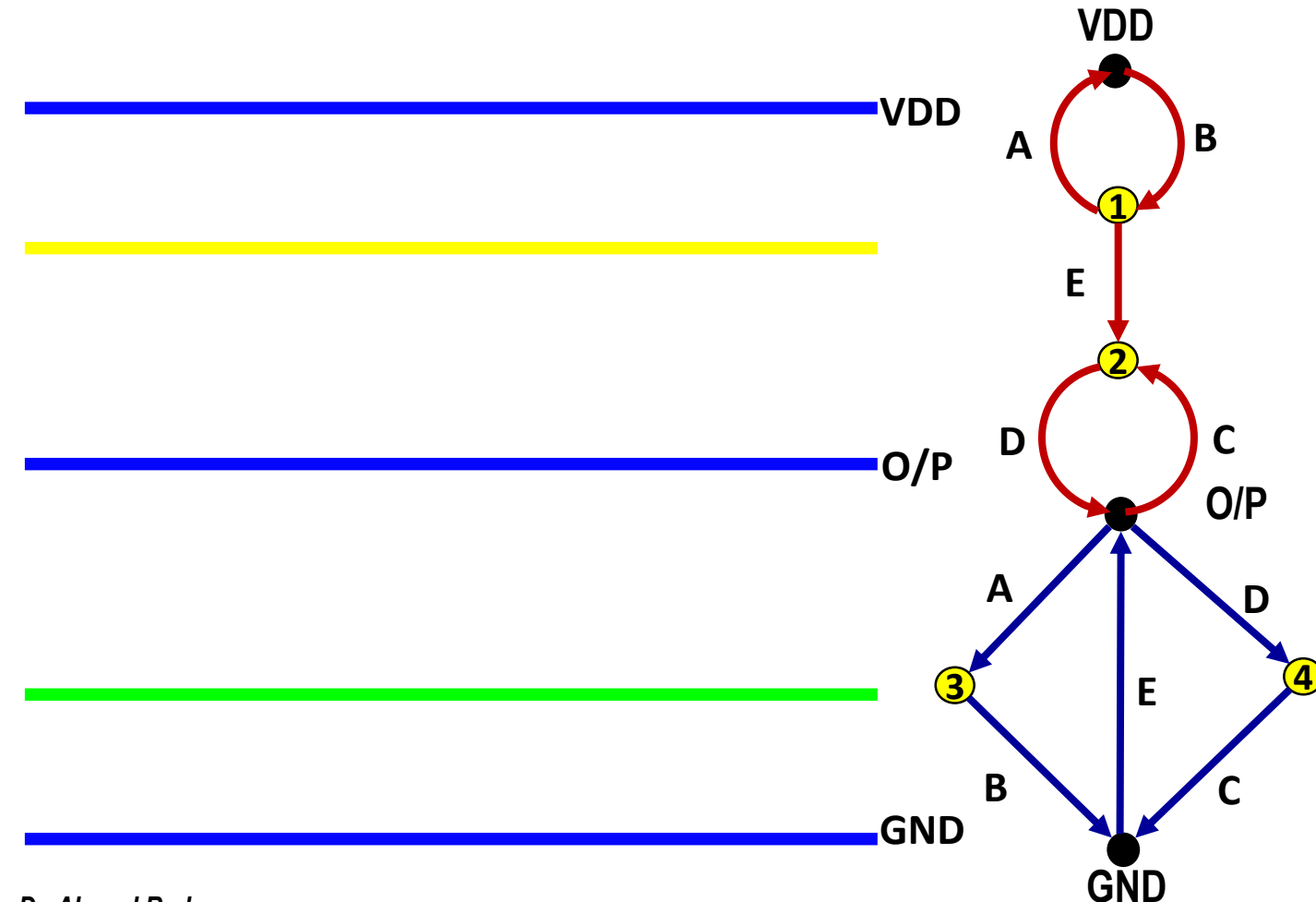
Exercise_02



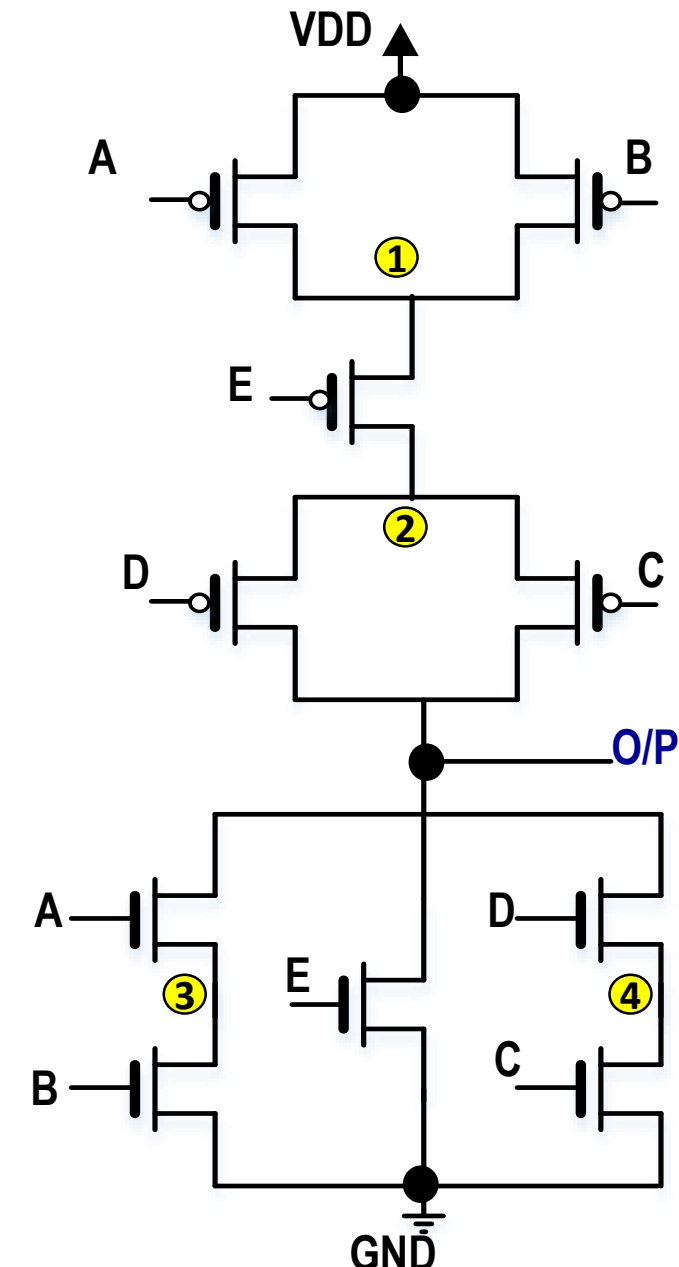
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□ Step5:- Draw the stick diagram

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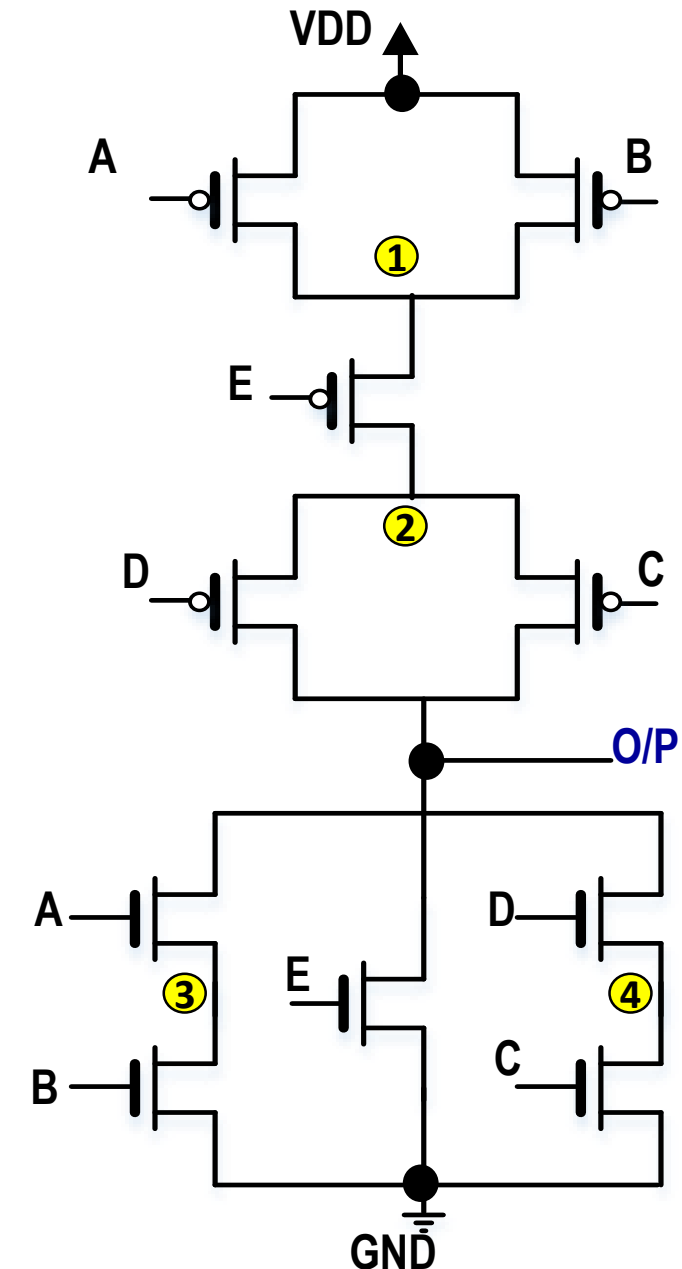
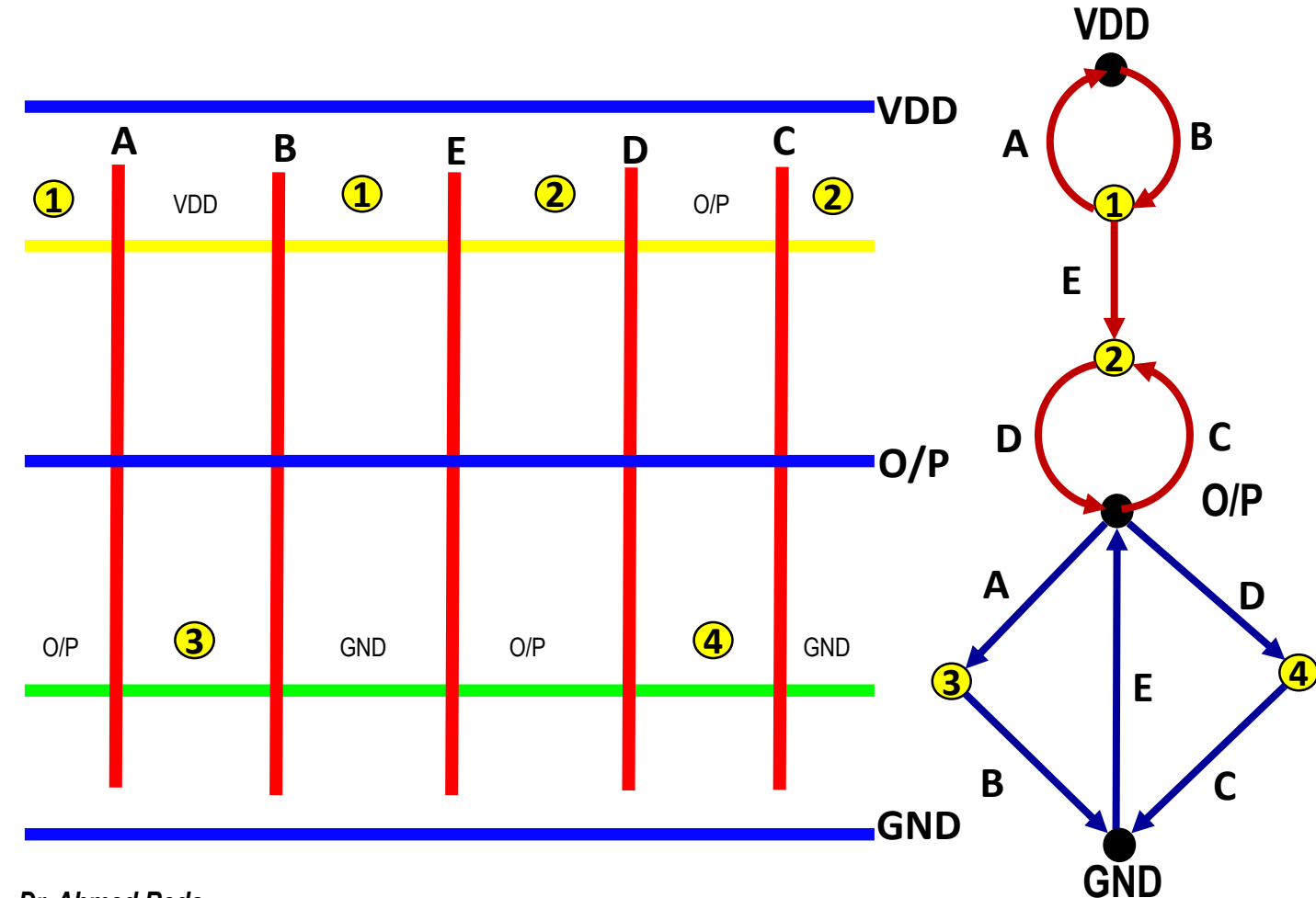


Exersice_02

□ Draw the stick diagram of the following circuit

□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow E \rightarrow D \rightarrow C$

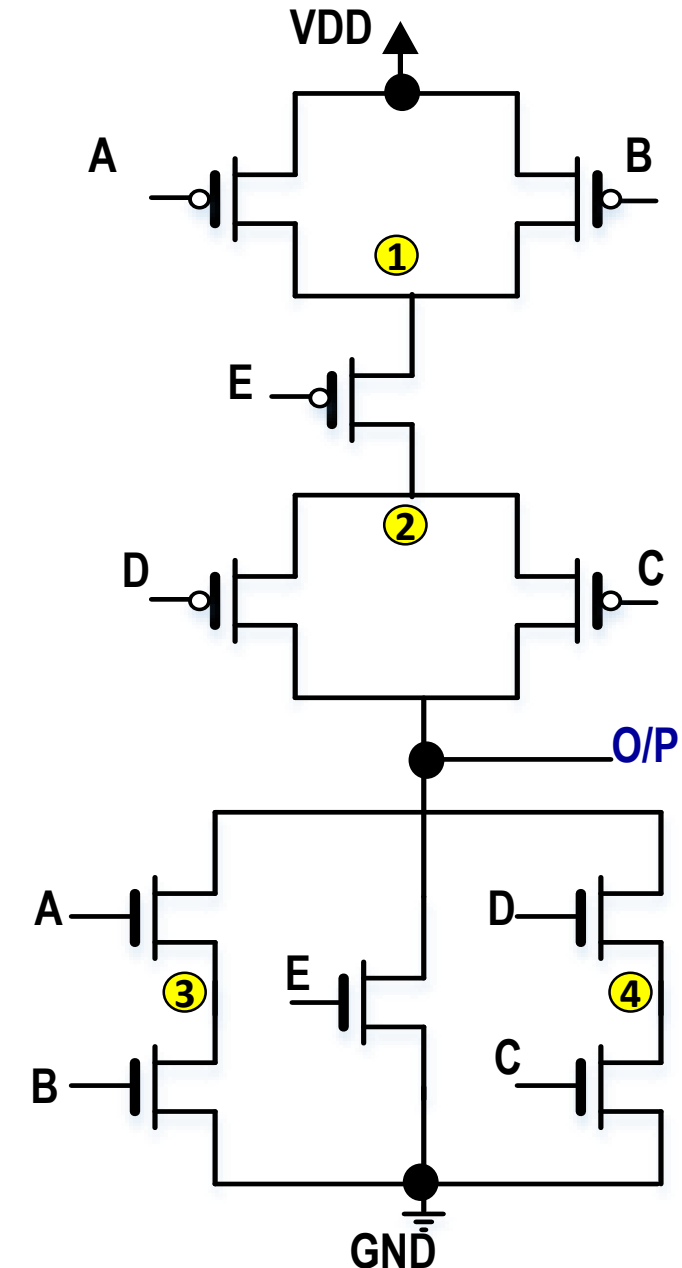
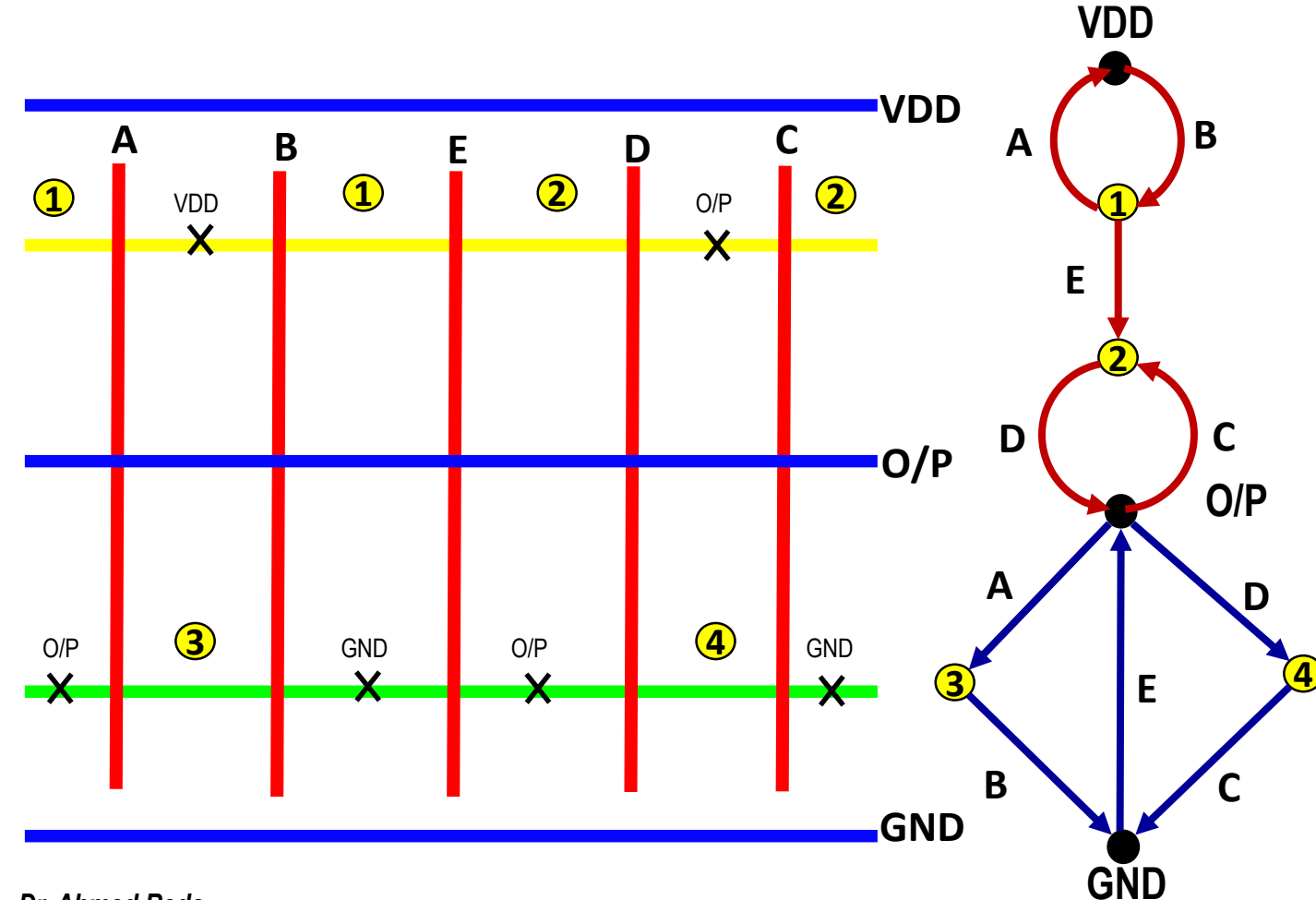


Exersice_02

□ Draw the stick diagram of the following circuit

□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow E \rightarrow D \rightarrow C$

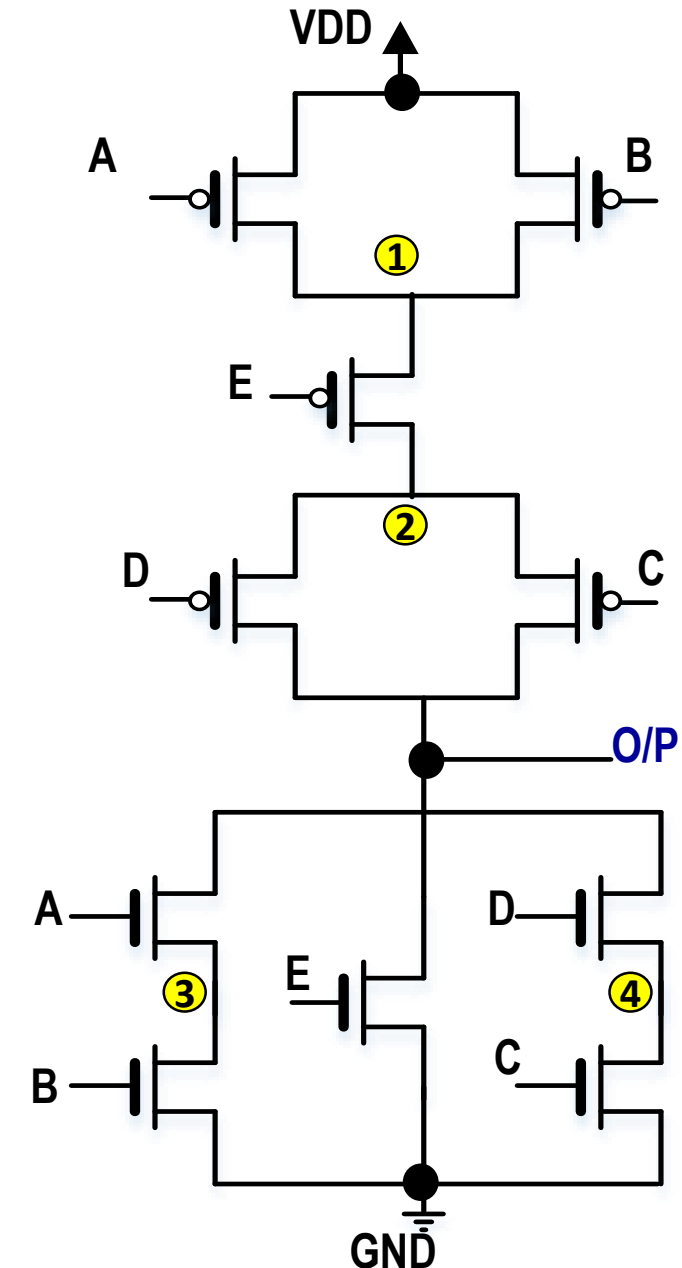
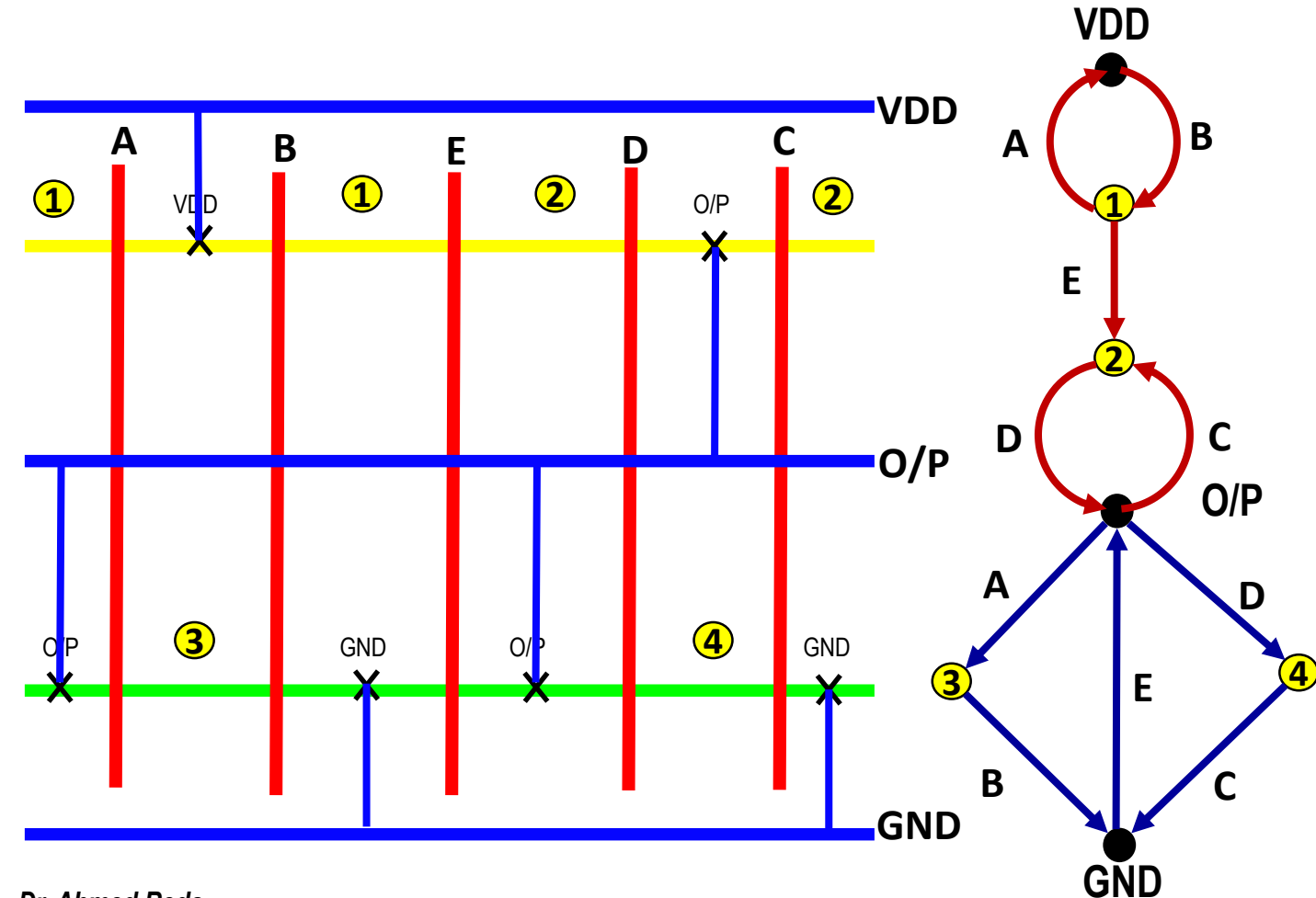


Exersice_02

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□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow E \rightarrow D \rightarrow C$

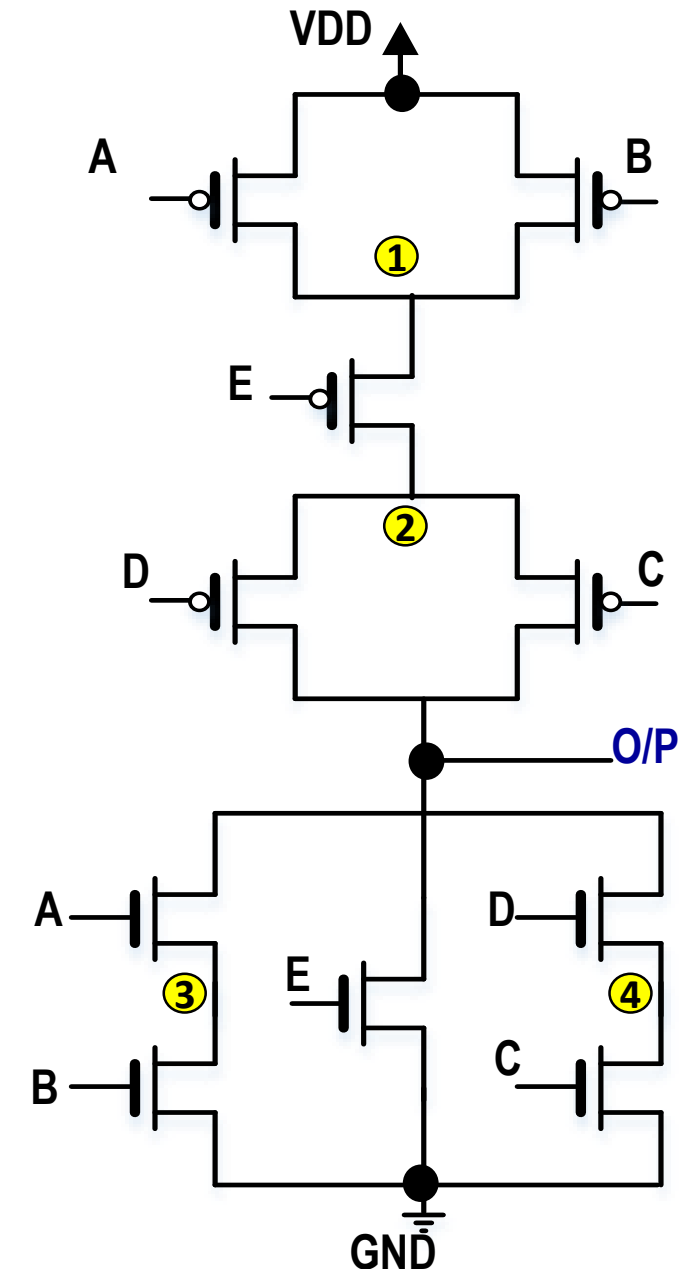
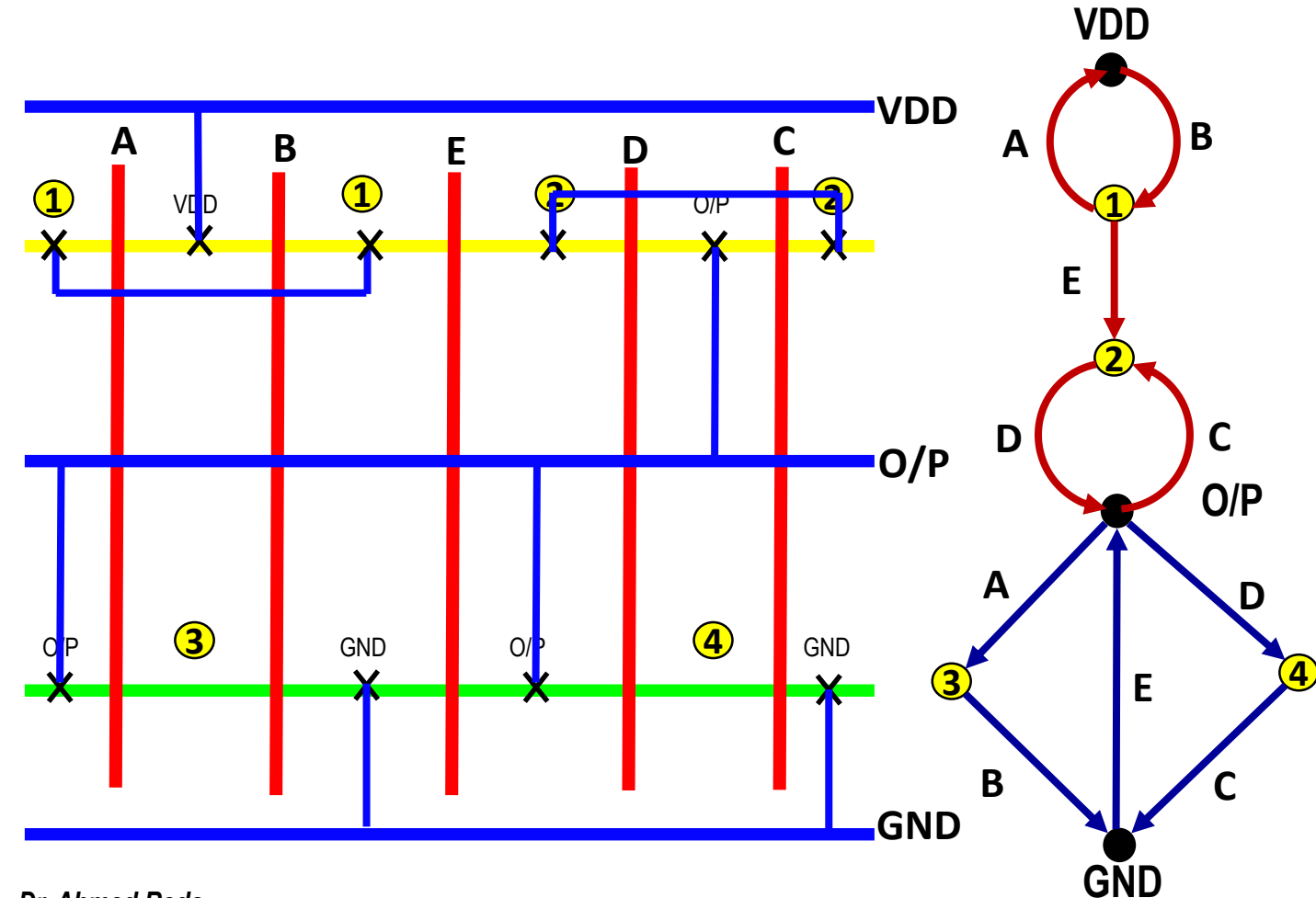


Exersice_02

□ Draw the stick diagram of the following circuit

□ Step5:- Draw the stick diagram

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Outlines

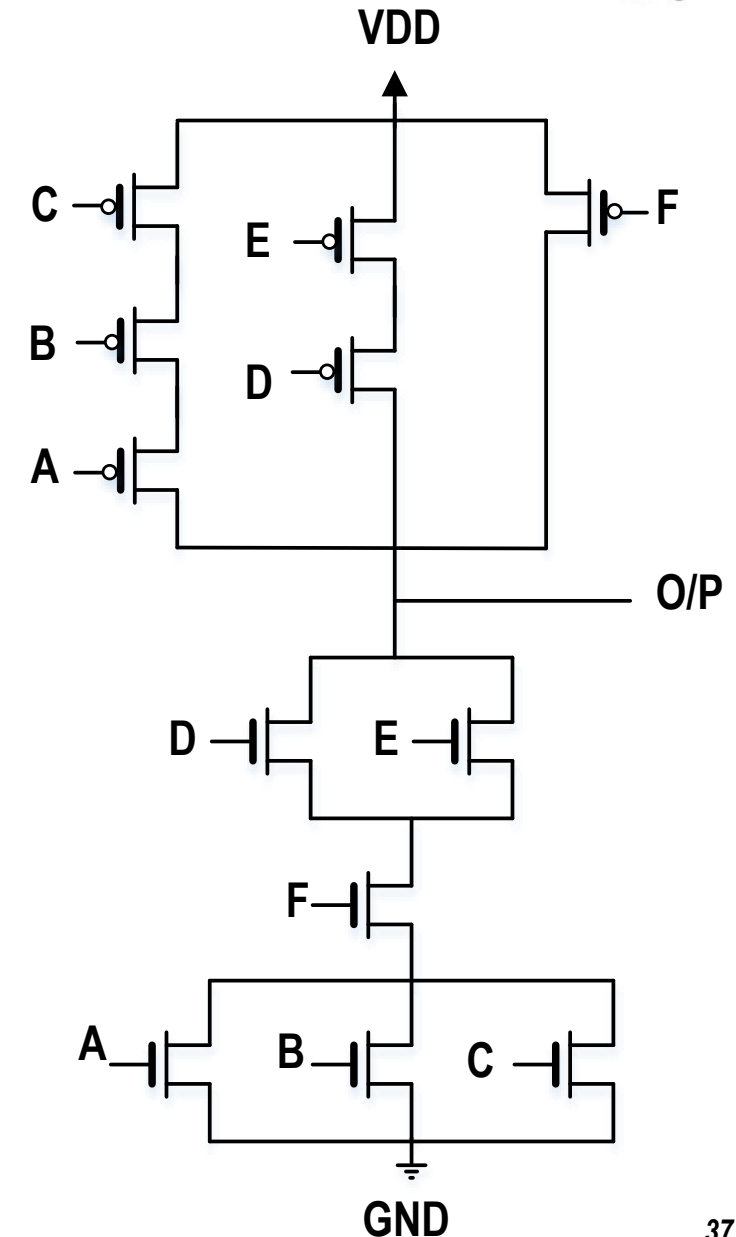


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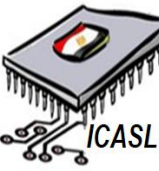
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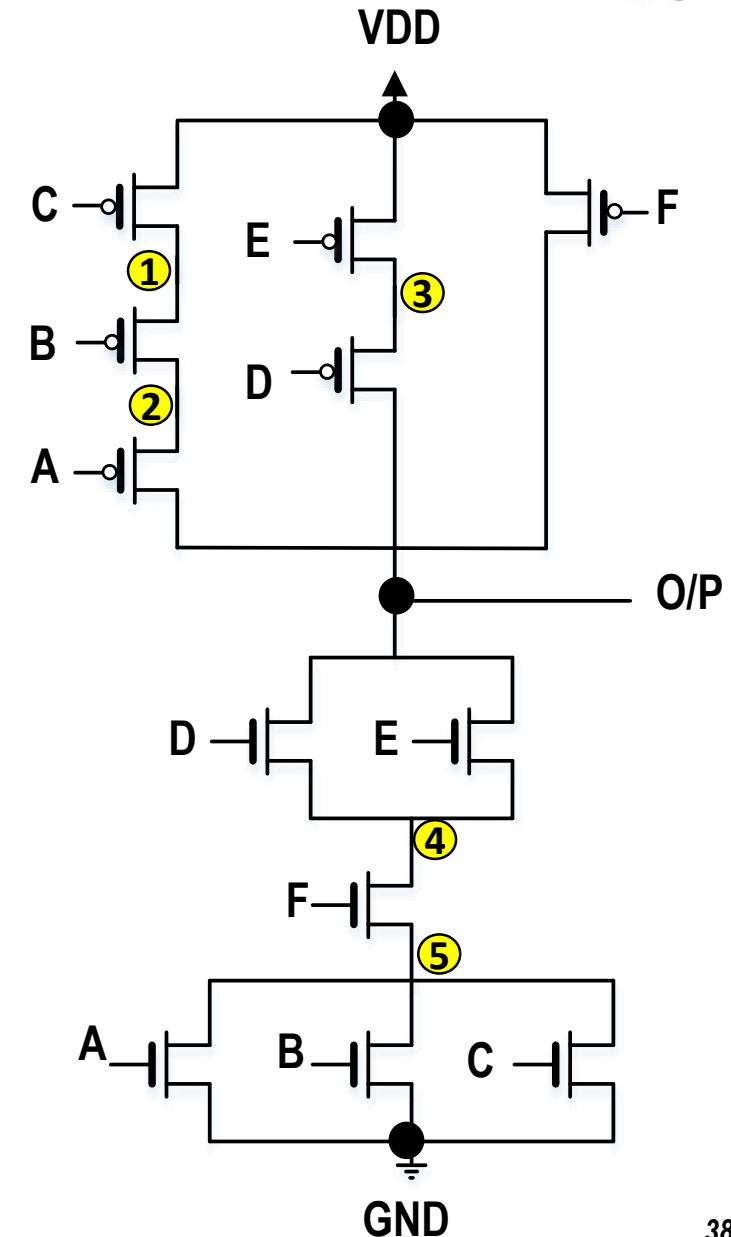
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Exercise_03



- ☐ Draw the stick diagram of the following circuit
- ☐ Step2:- Identify each connection between transistors only and give them a unique name.
 - ☐ Main Nodes: VDD, O/P, GND
 - ☐ Sub-Nodes: 1,2,3,4,5

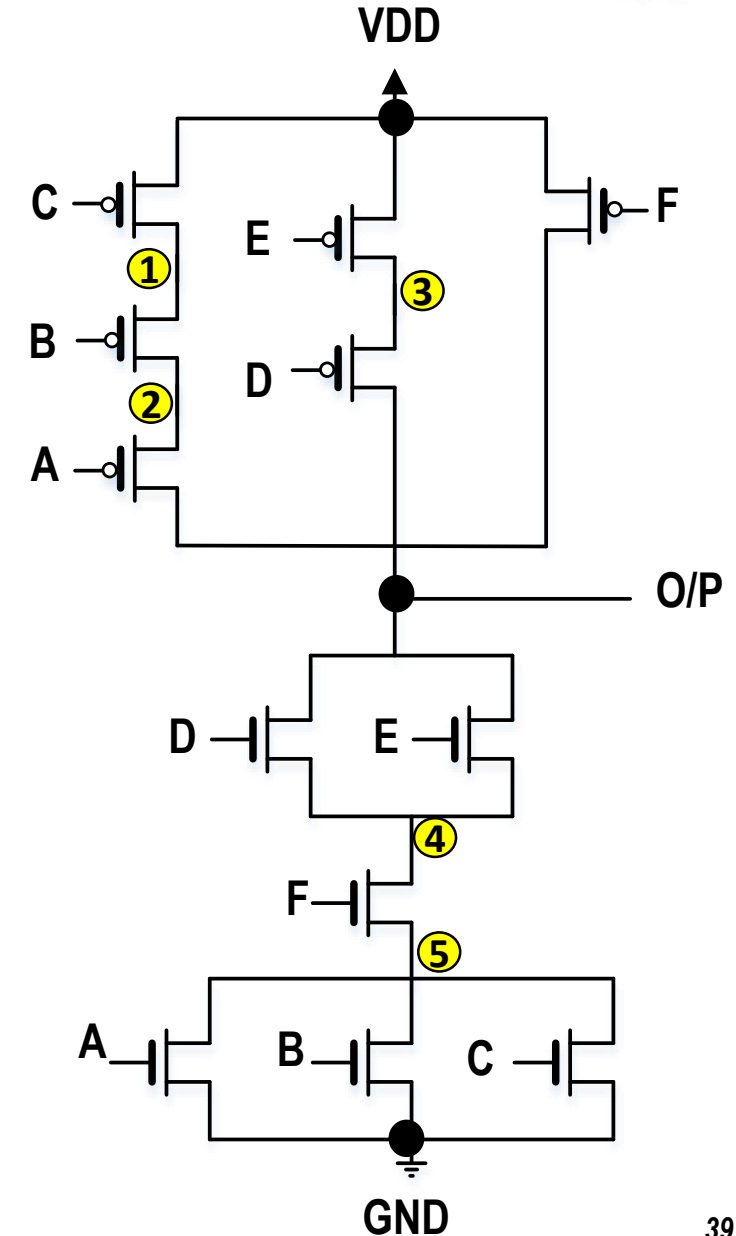
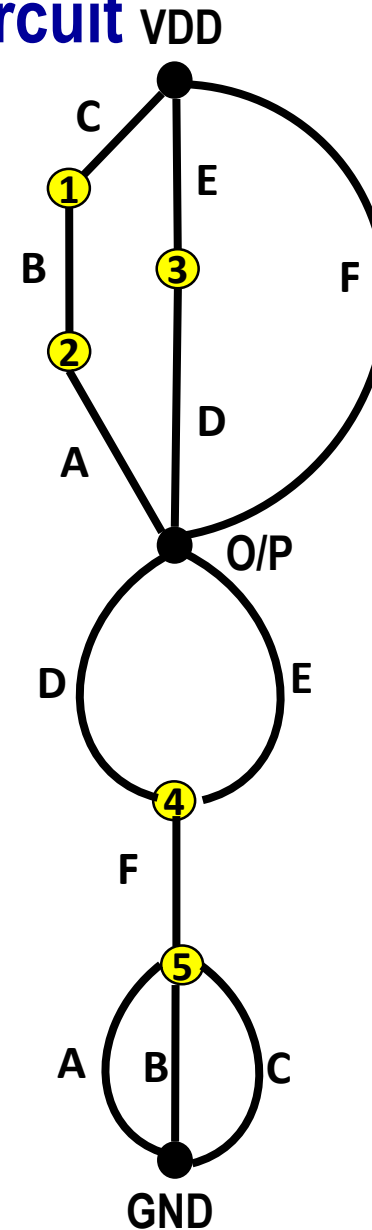


Exercise_03



□ Draw the stick diagram of the following circuit

Step3:- Draw the Euler diagram



Exercise_03

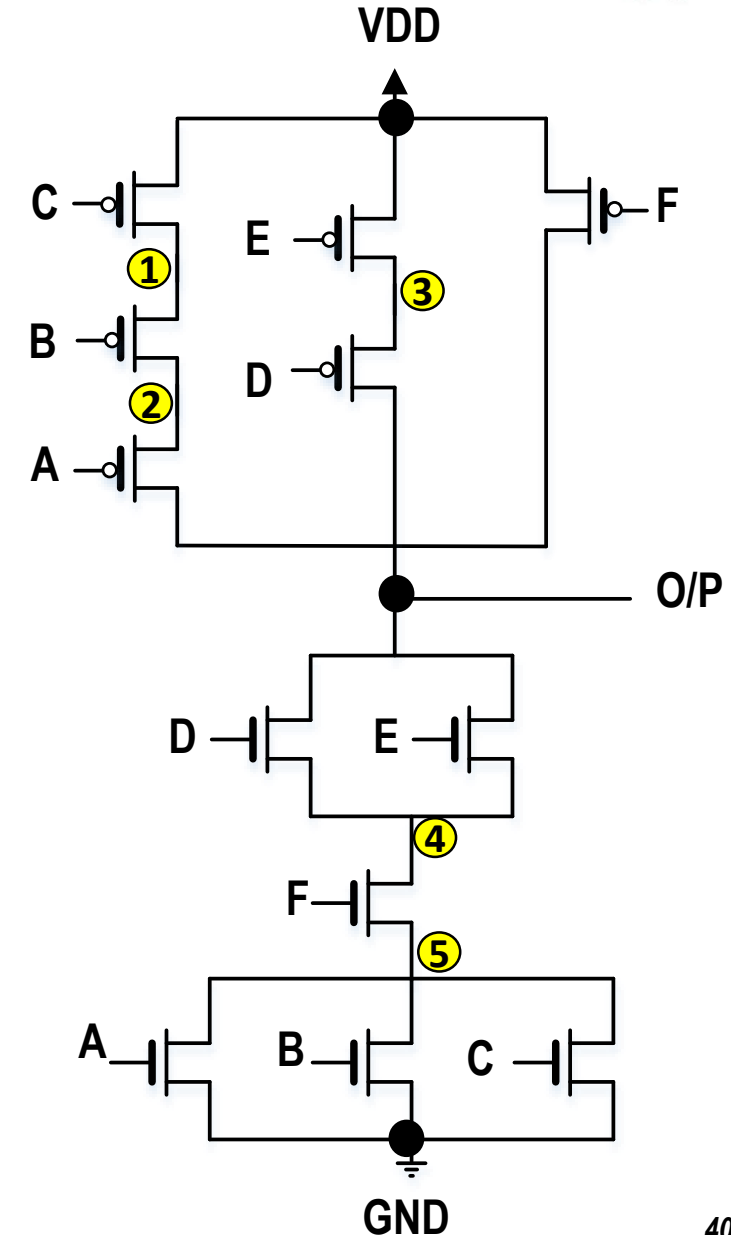
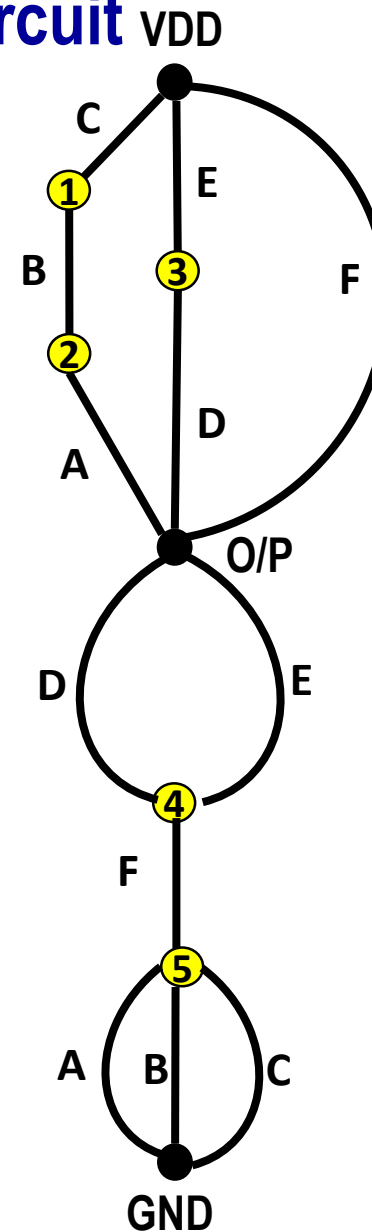


□ Draw the stick diagram of the following circuit

Step4:- Identify the Euler path

Condition of Euler Path

1. Nodes can be repeated but Paths cannot be repeated.
2. The path should be valid in both Pull-up and pull-down networks.



Exercise_03

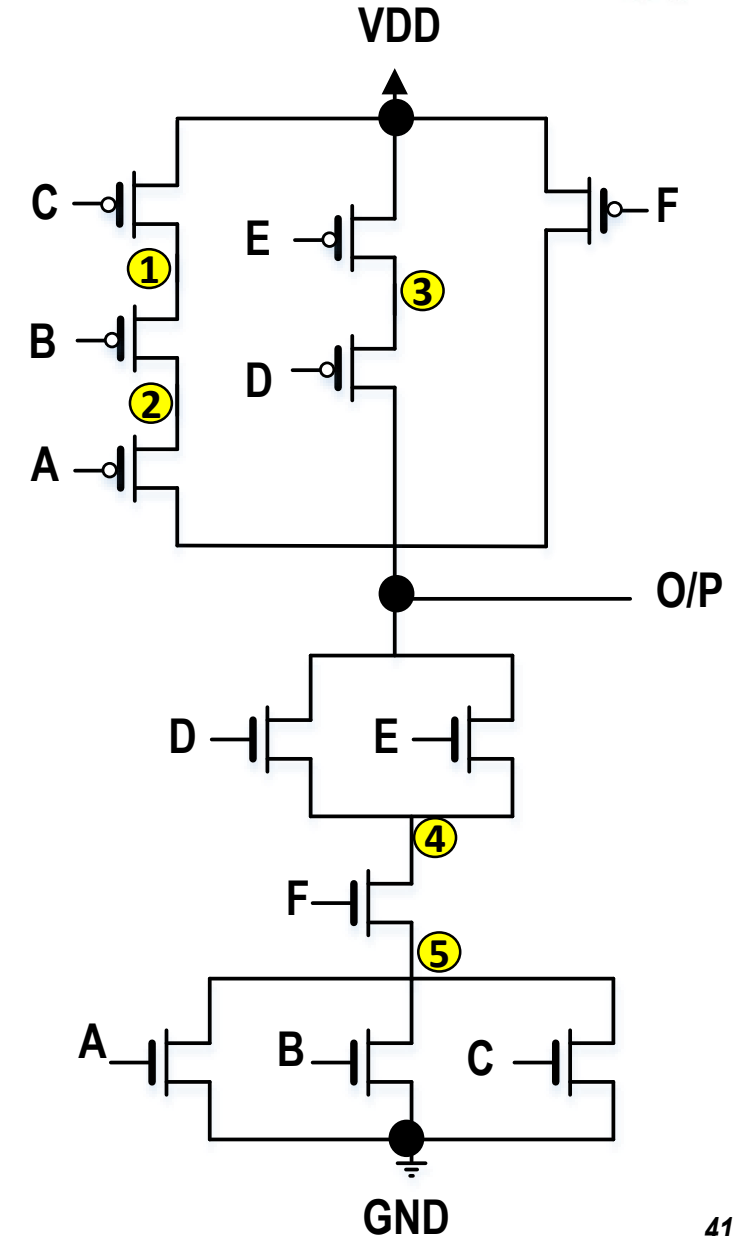
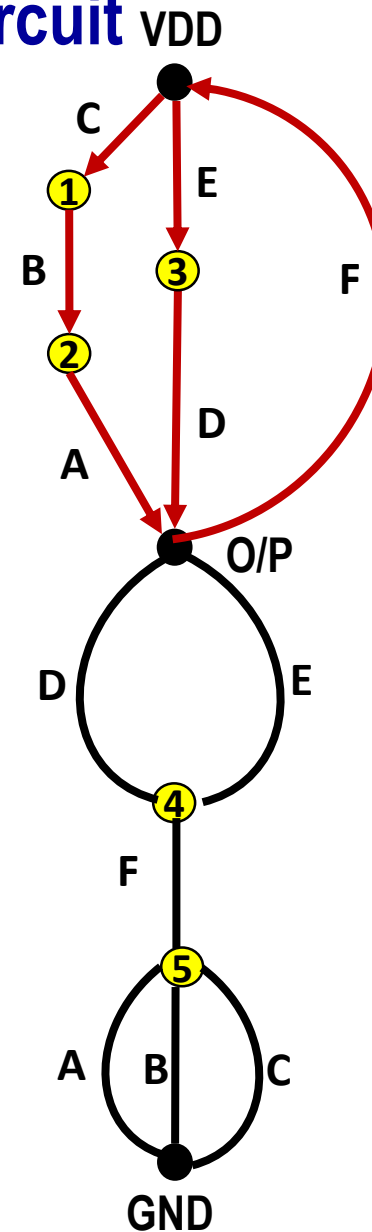


□ Draw the stick diagram of the following circuit

Step4:- Identify the Euler path

Euler path for Pull-up network is

C→B→A→F→E→D



Exersice_03



□ Draw the stick diagram of the following circuit

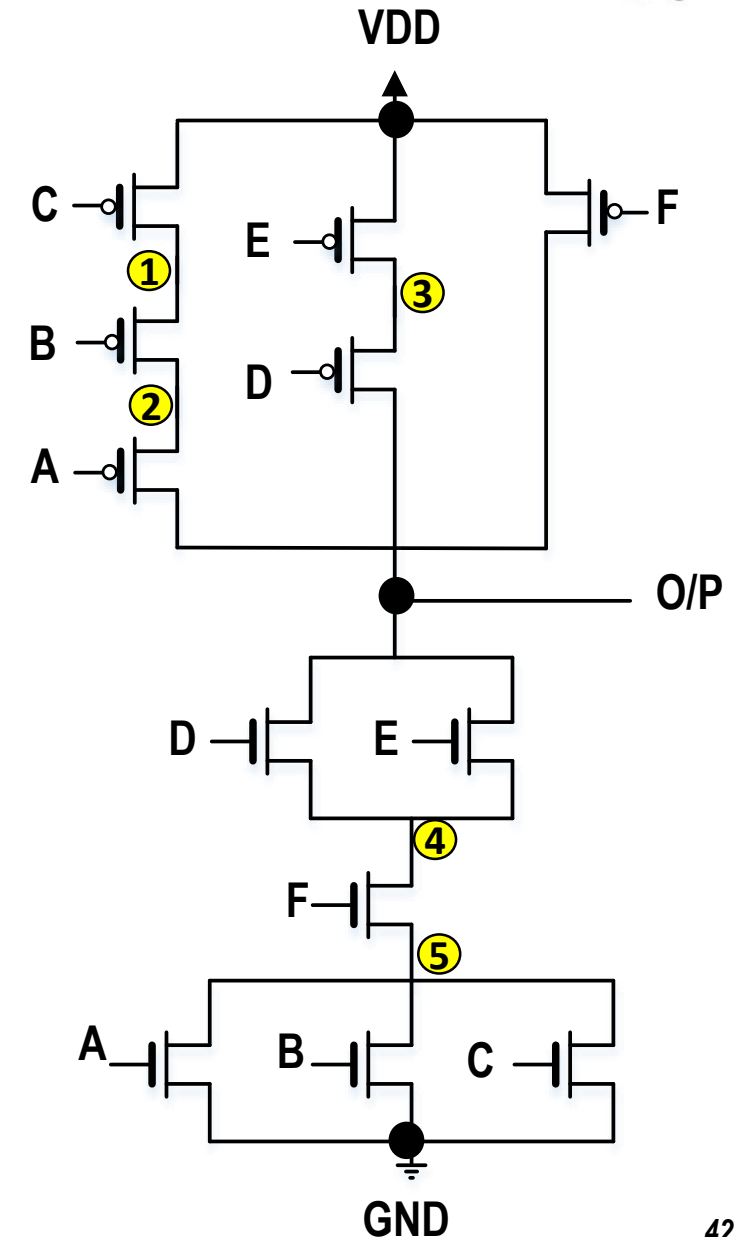
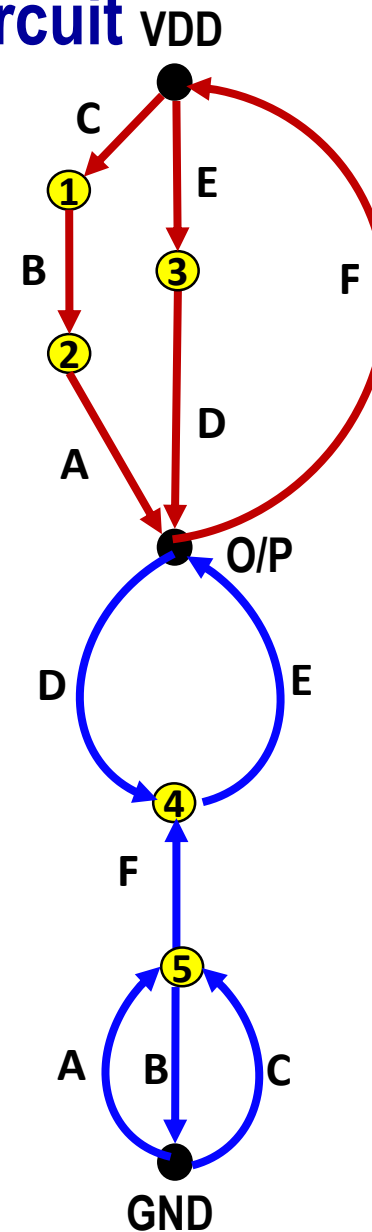
□ Step4:- Identify the Euler path

Euler path for Pull-up network is

C→B→A→F→E→D

Euler path for Pull-down network is

C→B→A→F→E→D



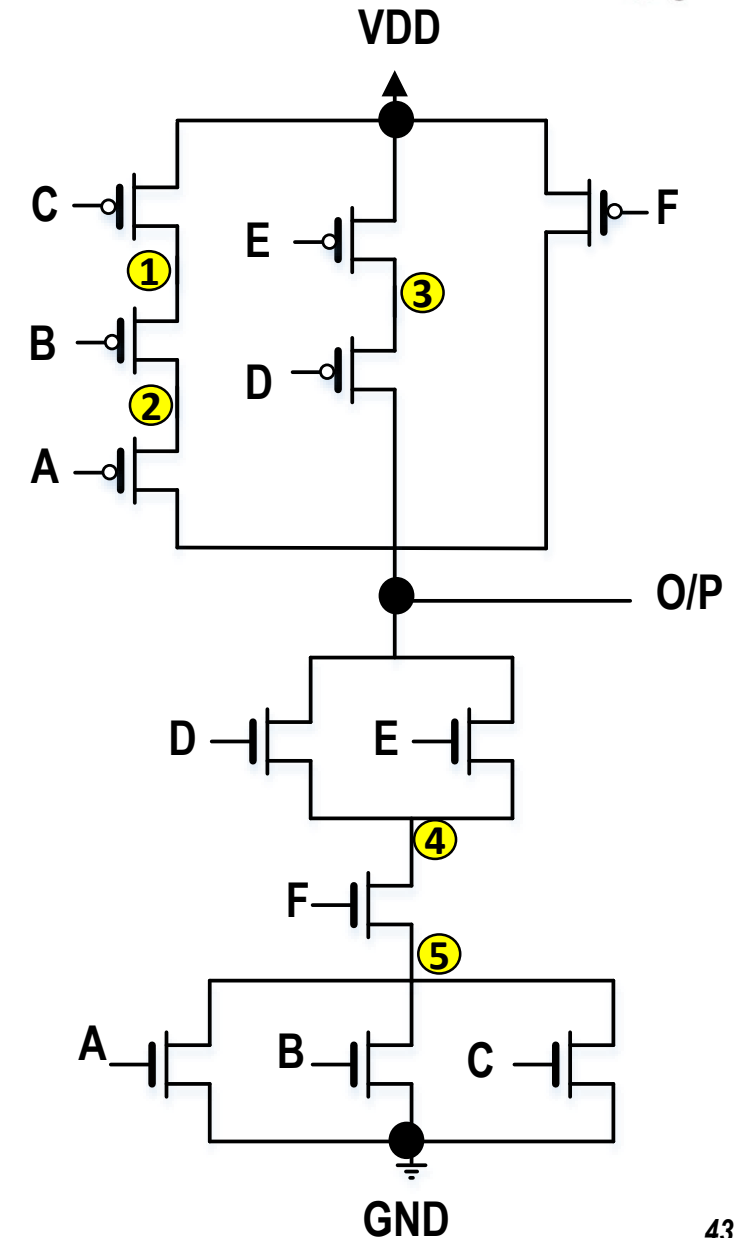
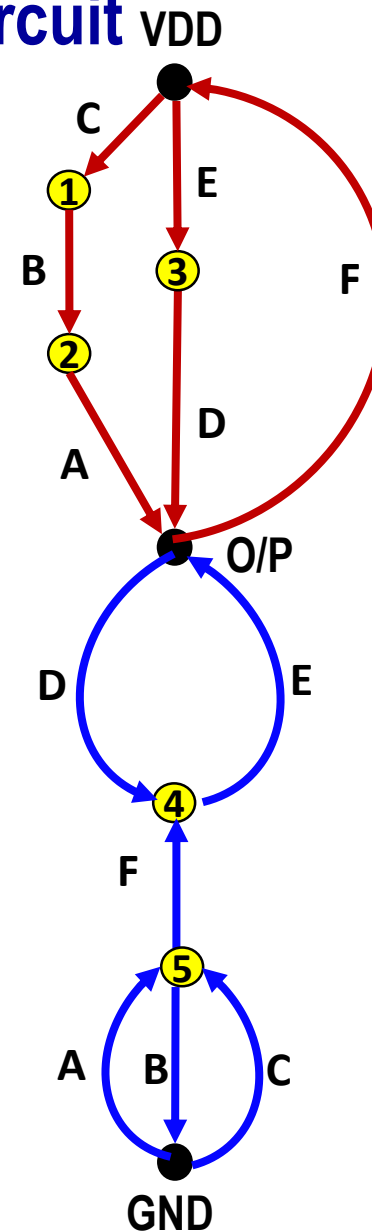
Exercise_03



□ Draw the stick diagram of the following circuit

Step5:- Draw the stick diagram

$C \rightarrow B \rightarrow A \rightarrow F \rightarrow E \rightarrow D$ (in order)



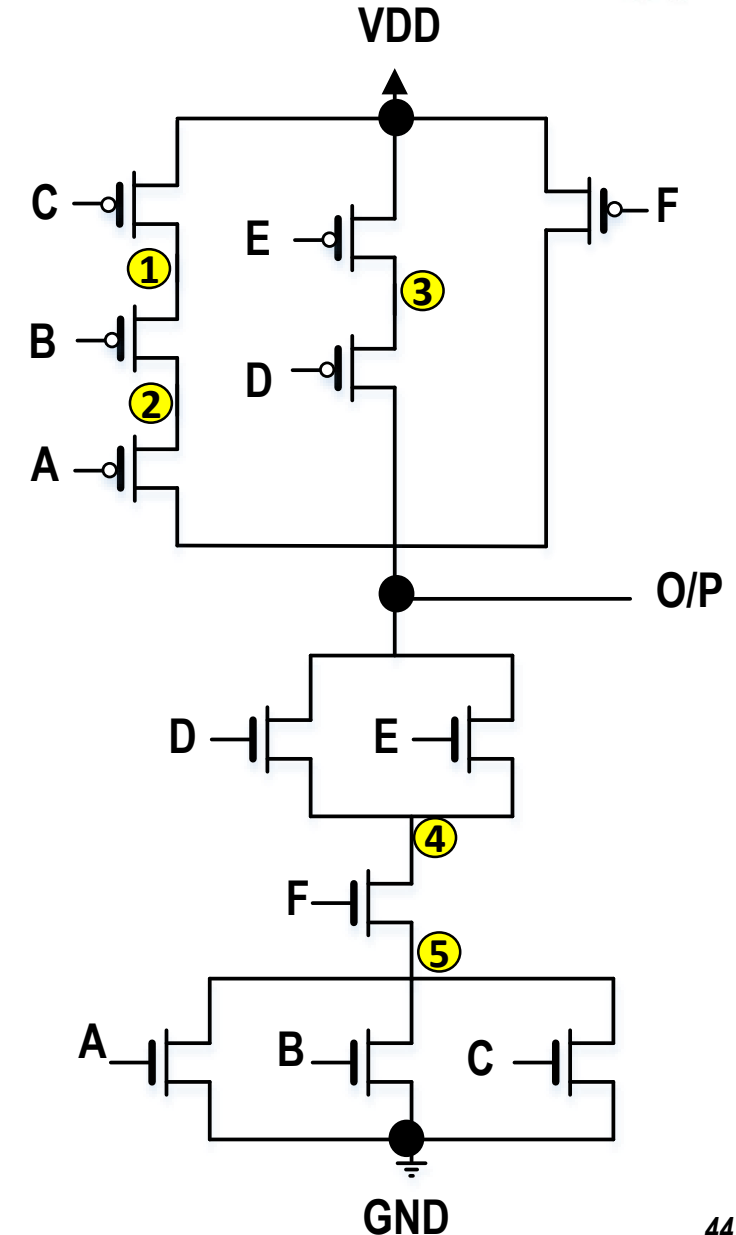
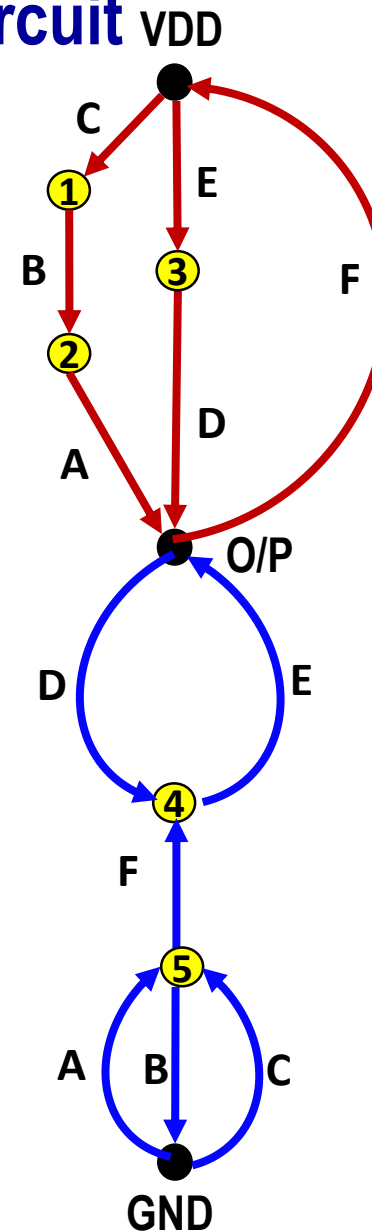
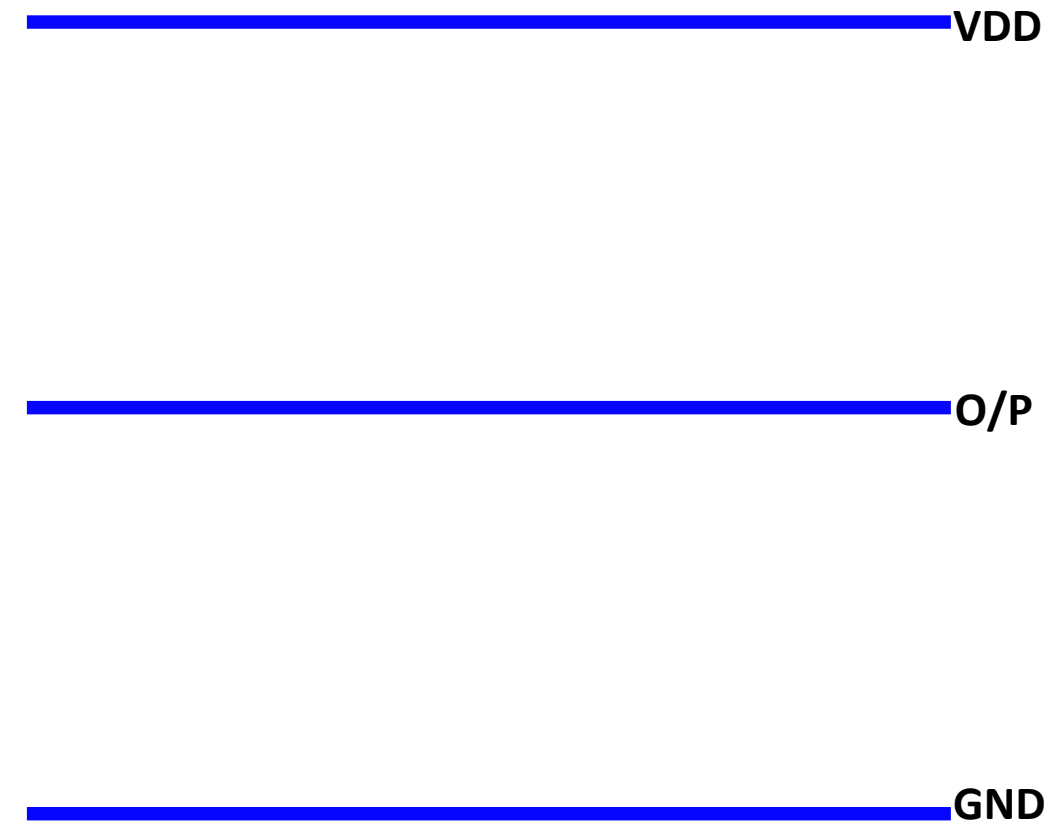
Exersice_03



□ Draw the stick diagram of the following circuit

Step5:- Draw the stick diagram

$C \rightarrow B \rightarrow A \rightarrow F \rightarrow E \rightarrow D$ (in order)



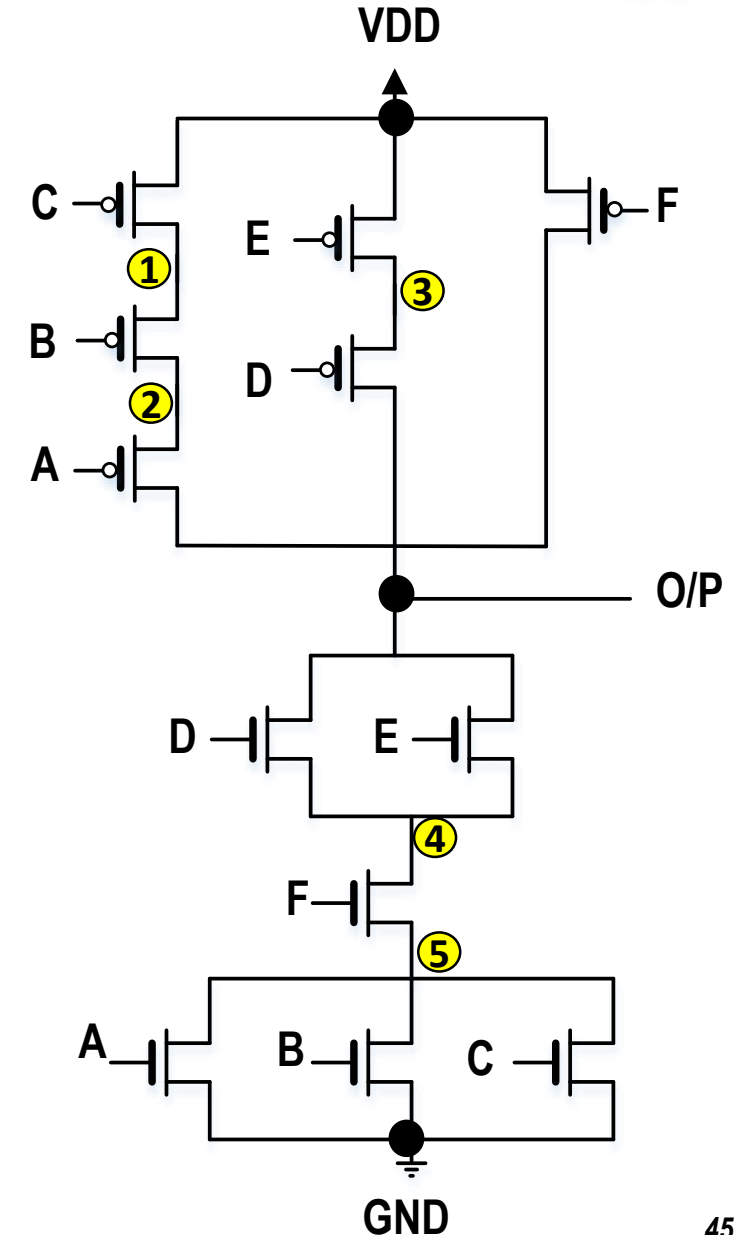
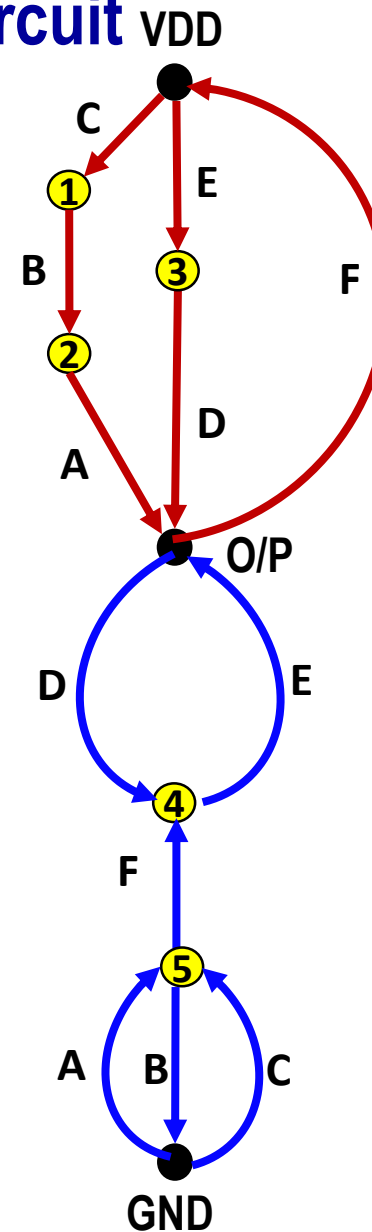
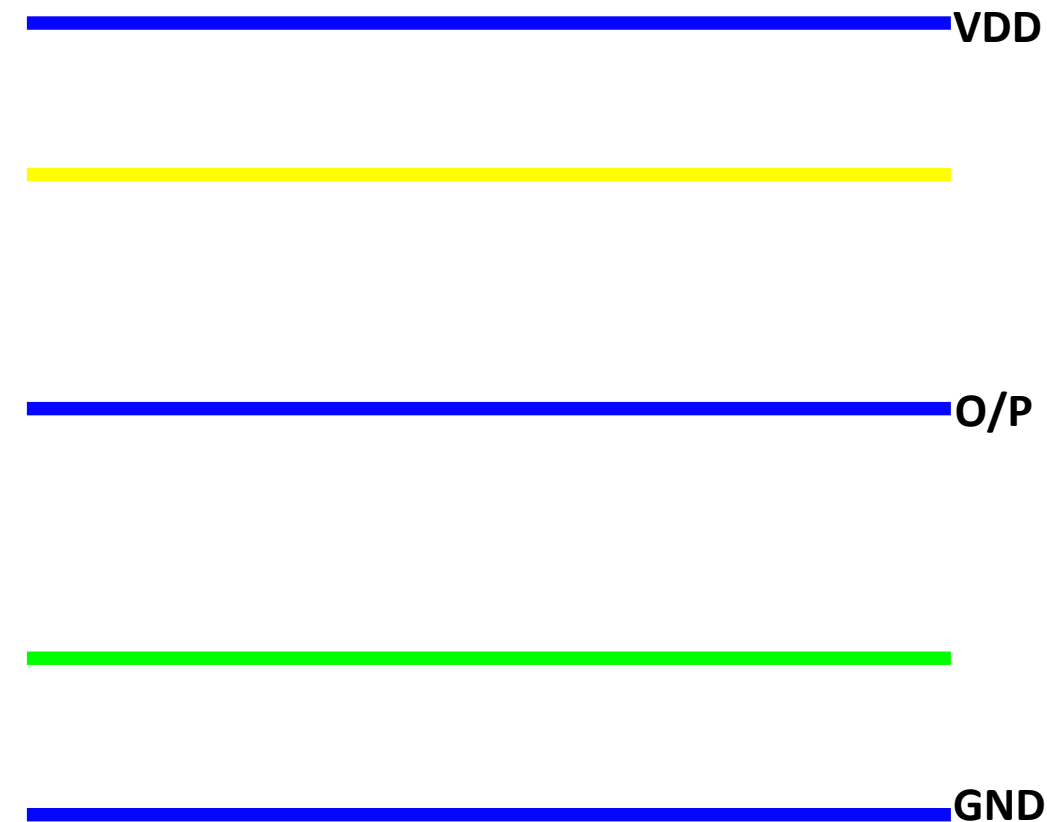
Exercise_03



□ Draw the stick diagram of the following circuit

Step5:- Draw the stick diagram

$C \rightarrow B \rightarrow A \rightarrow F \rightarrow E \rightarrow D$ (in order)



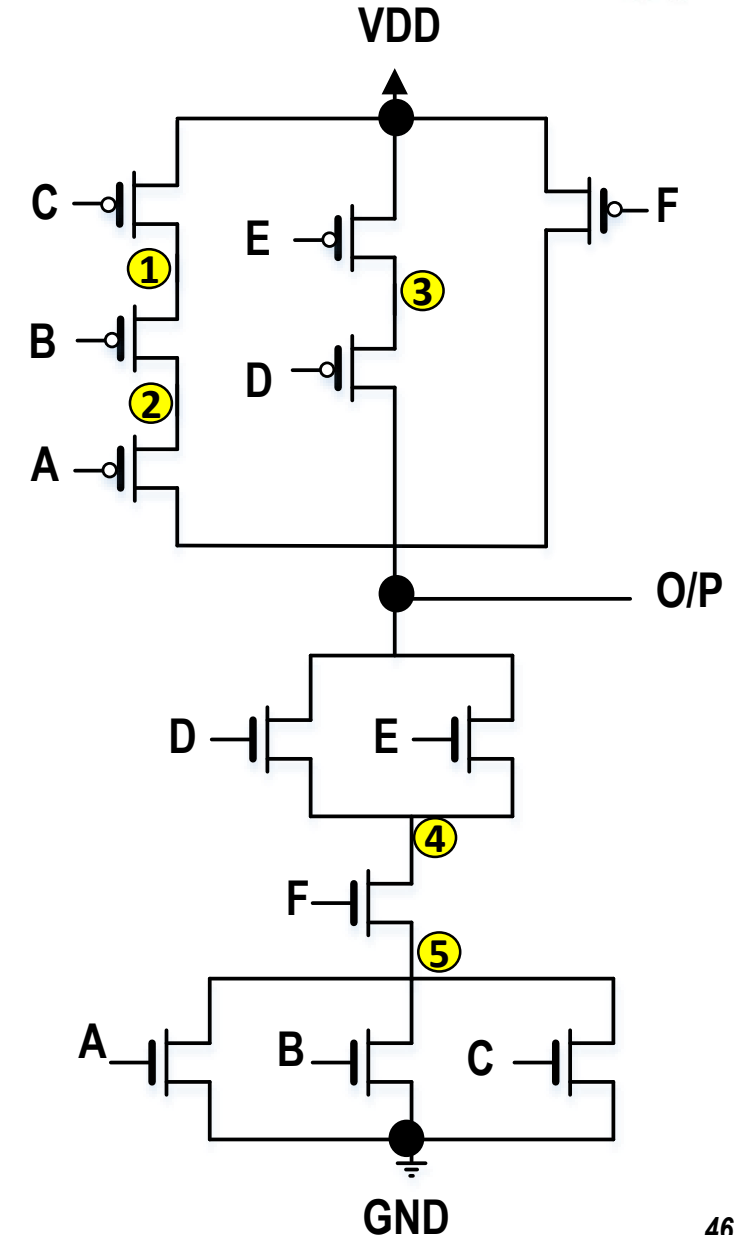
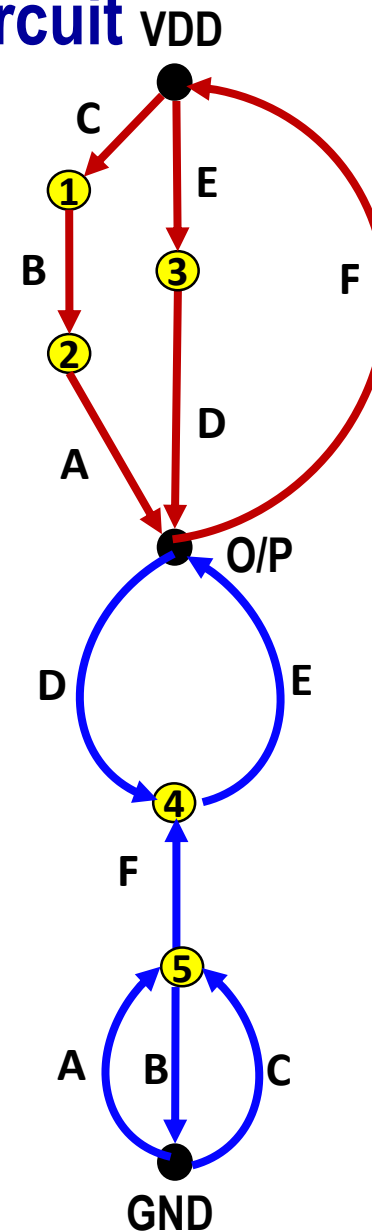
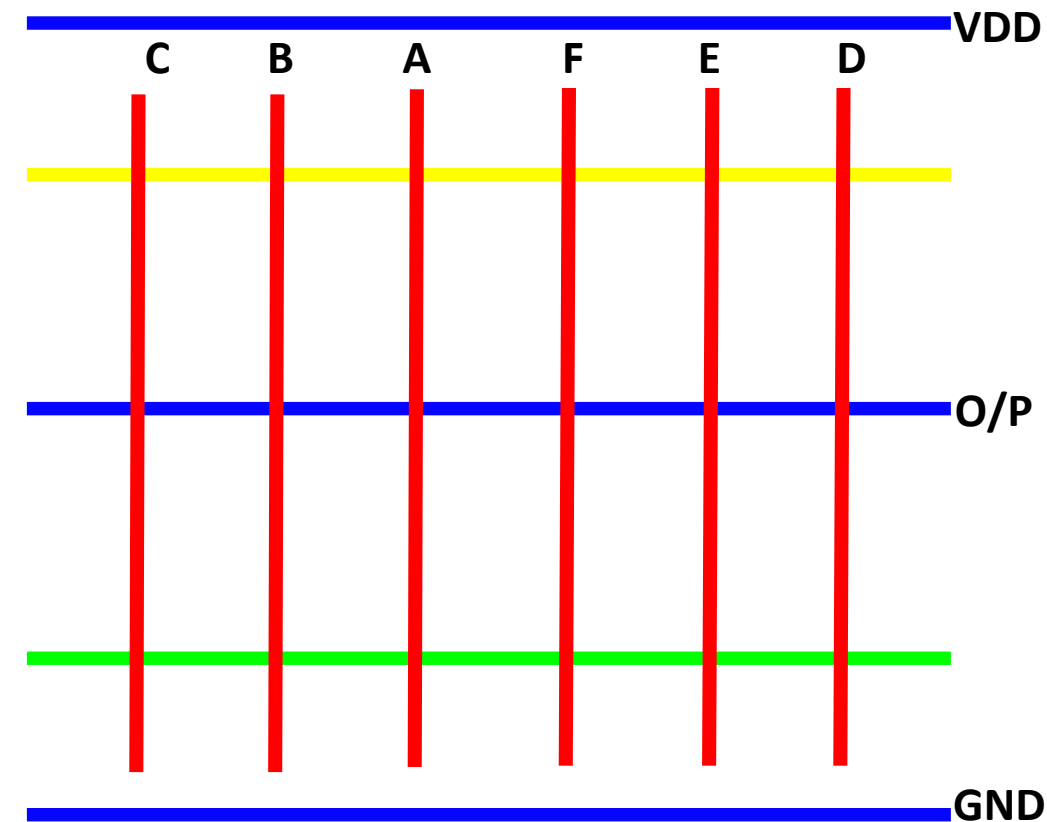
Exercise_03



□ Draw the stick diagram of the following circuit

Step5:- Draw the stick diagram

$C \rightarrow B \rightarrow A \rightarrow F \rightarrow E \rightarrow D$ (in order)



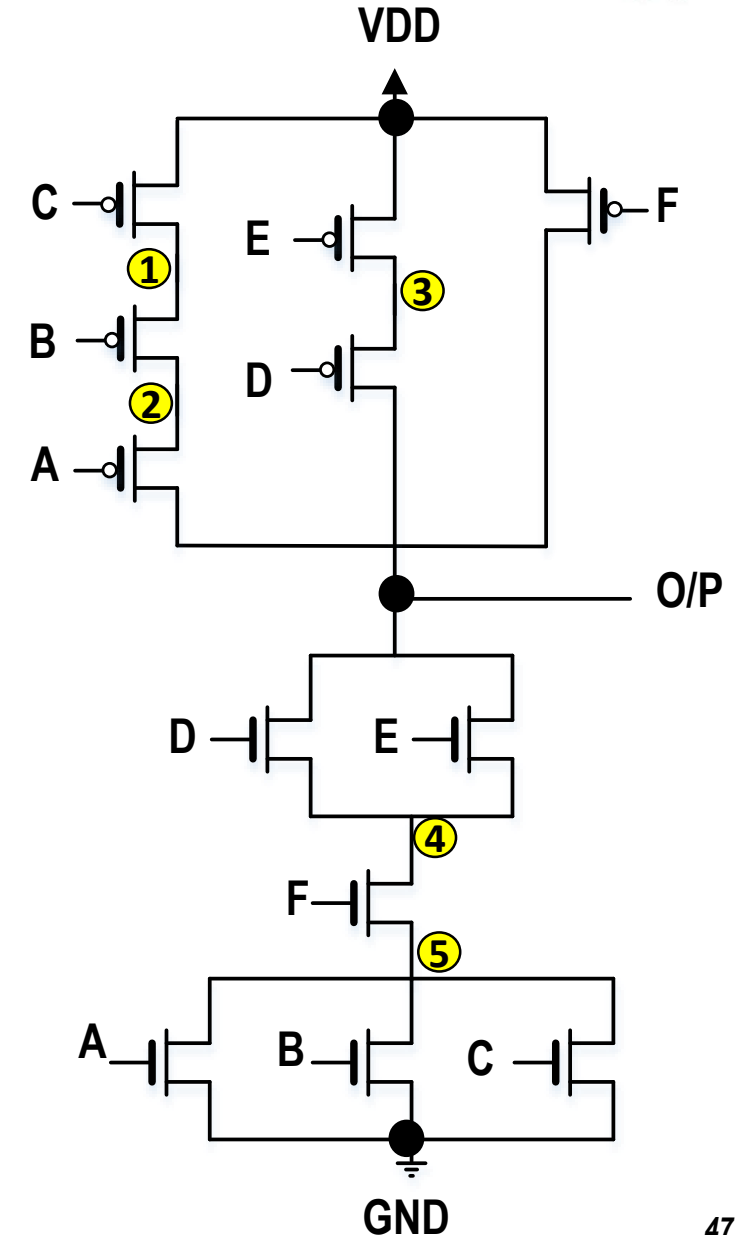
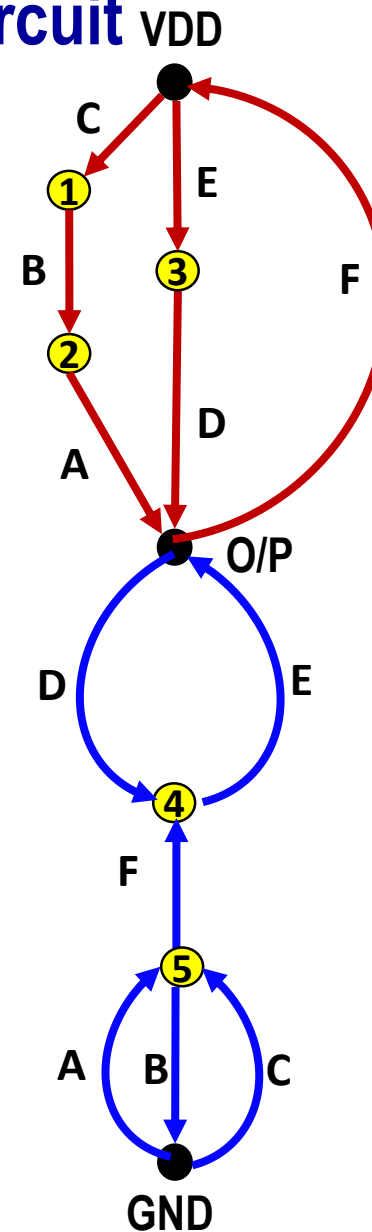
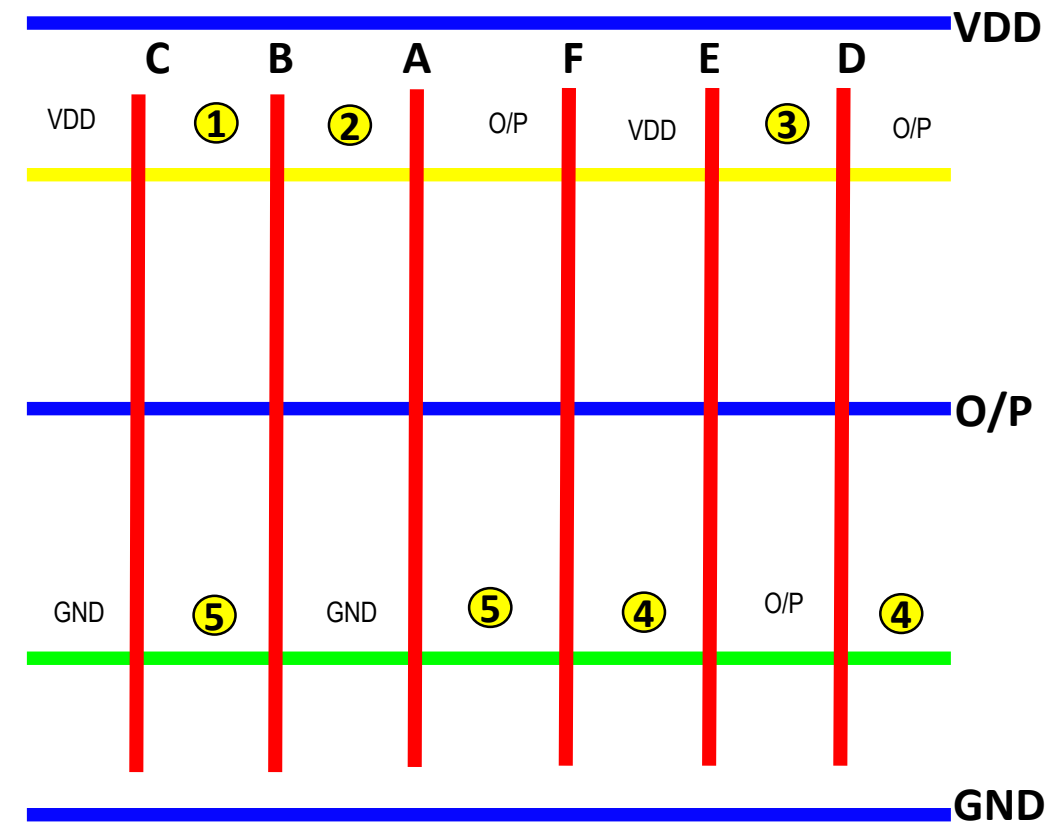
Exercise_03



□ Draw the stick diagram of the following circuit

Step5:- Draw the stick diagram

$C \rightarrow B \rightarrow A \rightarrow F \rightarrow E \rightarrow D$ (in order)



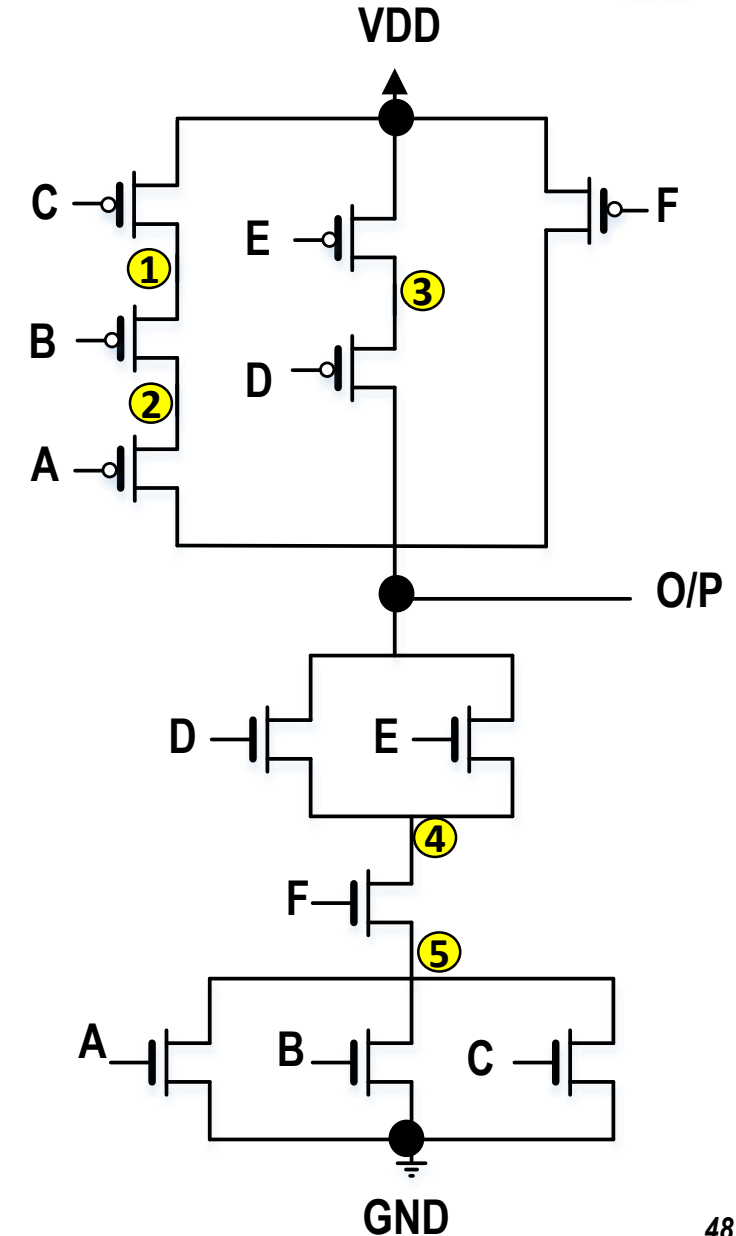
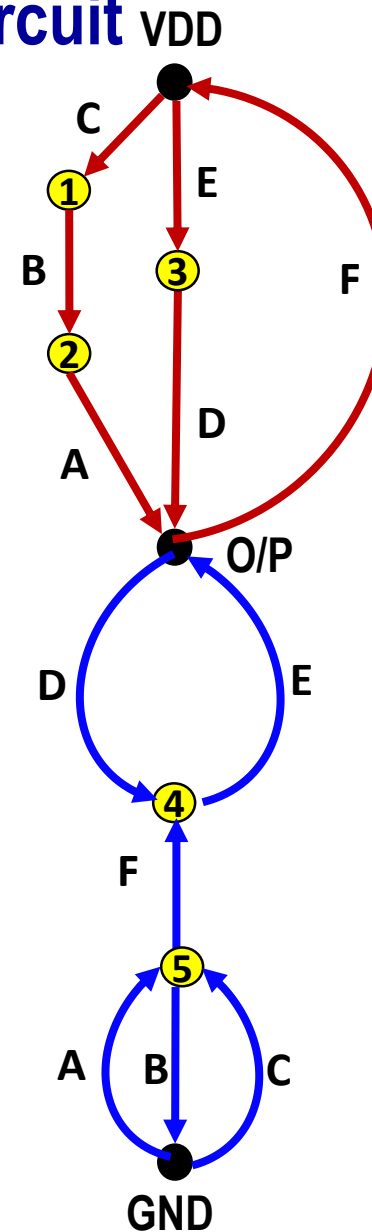
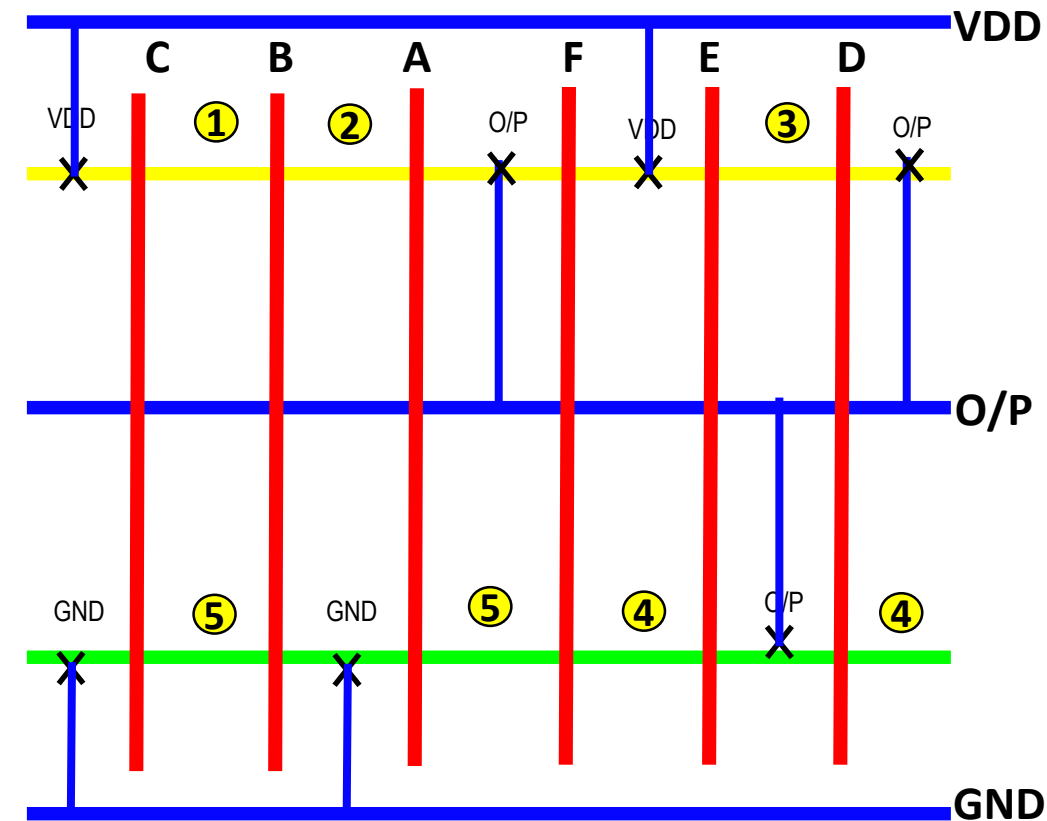
Exersice_03



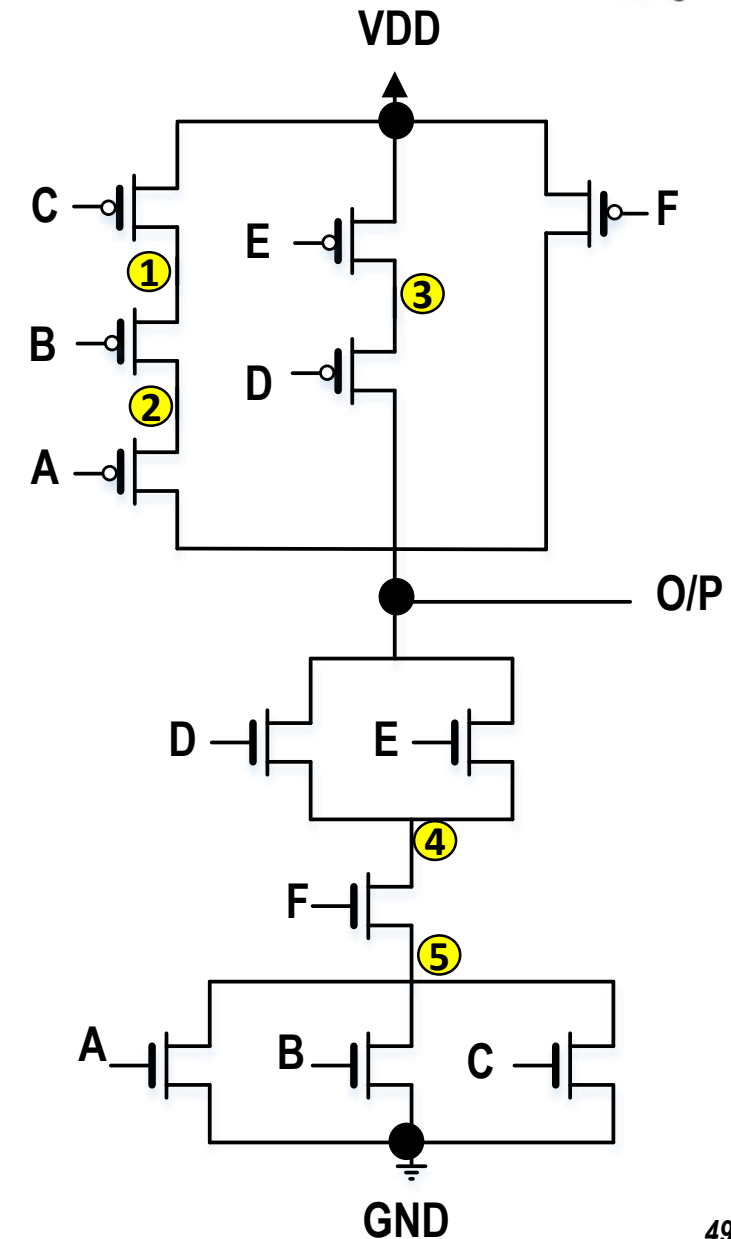
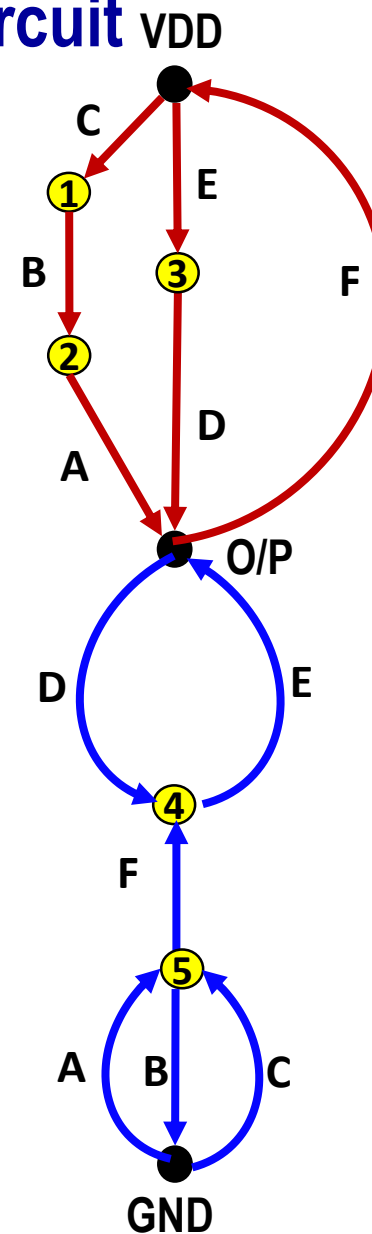
□ Draw the stick diagram of the following circuit

Step5:- Draw the stick diagram

$C \rightarrow B \rightarrow A \rightarrow F \rightarrow E \rightarrow D$ (in order)



Step5:- Draw the stick diagram



Thanks!